Foundations presenter script

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78 slides / 165 teaching minutes / two worked exercises

A speaking and teaching companion for all 78 foundations slides. Follow the existing 165-minute route, including the six-minute exercise on slide 26 and the eight-minute exercise on slide 53. Add breaks outside this teaching time.

Use Say as the spoken explanation. Walk the image or concept using Show / explain, then use the quick question when it helps the room. Expected answers are presenter cues. The final sentence bridges to the next slide. The timings include explanation, visual walkthrough and brief discussion; the exercise timings include all group work.

This script follows the accepted deck and its 18 October 2026 evidence snapshot. It preserves the distinction between company disclosures, supplier reports, product announcements, research results and invented teaching examples. It makes no new experiment or model call.

Before presenting

Open the presentation and script in separate windows. Test presenter view and source links.

Ask learners to keep one real work task in mind. Revisit it in the closing discussion.

Name the task, denominator and scoring rule for every research number. Different tasks do not form a model ranking.

Keep exercise answers private until the debrief. Foundations needs no notebook or model account.

Open the foundations slides

Open presenter view

Presenter script source in the repository

Teaching route

1. Agents, tools and workflows: slides 1-8, 00:00:00-00:11:00.
2. What power-sector organizations report: slides 9-17, 00:11:00-00:28:30.
3. Operator tasks and evidence checks: slides 18-26, 00:28:30-00:48:00.
4. Research foundations, LangGraph and PowerDAG: slides 27-42, 00:48:00-01:19:30.
5. Simulation feedback and valid reuse: slides 43-53, 01:19:30-01:53:00.
6. Power-system benchmarks and failure modes: slides 54-65, 01:53:00-02:19:30.
7. Statistics, integration and evaluation: slides 66-75, 02:19:30-02:40:00.
8. Prepare for the three-case workshop: slides 76-78, 02:40:00-02:45:00.

Exercise pacing

Slide 26: six minutes. Inspect alone for one minute, work in pairs for two, share for two and debrief for one.

Slide 53: eight minutes. Think alone for two minutes, compare in groups for two, discuss for two and debrief for two.

The printed script gives one page per slide. Full author, title, publication details and source URLs follow the glossary. Source IDs on each teaching page link to that reference list.

Full reading guide and visual provenance

Follow-on hands-on tutorial

1. Agents, tools and workflows

Slide 1. Agentic AI for Utility Operators

00:00:00-00:00:30 / 0.5 minutes

Say

Welcome. I am Alaa Selim, from MIT/Dartmouth, presenting with Mohamed Chehada from UT Austin. Our question is how an AI agent can help a power engineer or operator complete a useful, checkable task. We will define the basic ideas, examine industry reports, work through power-system research and learn how to judge the evidence. Two short exercises will let you make the checks yourself. The separate hands-on tutorial then applies these ideas to voltage support, restoration and cyber response.

Show / explain

Introduce the presenters and the session route. Ask learners to keep one task from their own work in mind throughout the session.

Ask

Which task would you like an assistant to help you complete?

Expected answer

Accept a specific task such as checking study inputs, locating event records or reviewing an outage. Reuse one audience example later.

Transition

First, what do we mean by agentic AI?

Sources: [B01]. Full references follow the glossary.

Open slide 1 online

Slide 2. What is agentic AI?

00:00:30-00:02:00 / 1.5 minutes

Say

We use agentic AI to describe a system that works toward a goal, reads the results of its actions and adjusts its next step. One agent can do this. Imagine asking whether a previous voltage study still applies after a switch changes position. The system may inspect the event, identify the study case, run an allowed check and ask for missing evidence. The important question is who chooses those steps and what feedback changes the choice. Definitions vary across the literature. Anthropic uses agentic systems to include both programmed workflows and model-directed agents. We will keep that source's terminology explicit rather than present one universal definition.

Show / explain

Trace each NVIDIA interaction pattern from the request through the model to tools or other agents. Point out that the same model can appear in more than one arrangement. The figure illustrates patterns rather than a required hierarchy.

Ask

Does an agentic system need several agents?

Expected answer

No. A single agent with appropriate feedback and tools can carry out a bounded task.

Transition

We should first separate content generation from choosing actions.

Sources: [DEF03], [DEF02], [NV01]. Full references follow the glossary.

Open slide 2 online

Slide 3. Generative AI creates content

00:02:00-00:03:30 / 1.5 minutes

Say

Generative AI produces content such as text, images or audio. A large language model, or LLM, handles language. In our report example, selected records and a request enter the model and a draft comes out. That can be useful on its own. An agent system can use the same model repeatedly to decide which evidence to request next. These ideas overlap. They describe roles within a system, rather than three technologies that replace one another. A clear paragraph about voltage also needs a traceable numerical result behind it. Fluent wording tells us how readable a report is. Engineering evidence tells us whether the conclusion follows from the study.

Show / explain

Follow the diagram from records and request to the model and draft. Identify the engineering records as inputs. Ask where a numerical claim in the draft could be checked.

Ask

If a model writes a good study summary, has it necessarily acted as an agent?

Expected answer

No. A single generation call may create the summary. We need to inspect the surrounding process to identify decision making and tool use.

Transition

The classical definition of an agent helps us identify that process.

Sources: [GEN01], [DEF03]. Full references follow the glossary.

Open slide 3 online

Slide 4. What is an AI agent?

00:03:30-00:05:00 / 1.5 minutes

Say

An AI agent observes an environment and chooses actions that affect it. This definition predates large language models. In the textbook picture, observations arrive through sensors and actions leave through actuators. For a software study assistant, the observations may be case files, measurements and solver logs. Its actions may be requests to a database or a simulator. The decision rule uses what has been observed so far. That does not require a robot or a physical actuator. It also does not require unrestricted autonomy. We choose the task and permissions. A solver can remain responsible for electrical calculations while an agent decides which permitted study to request next.

Show / explain

Start with the environment, follow the sensor arrow into the agent, then the actuator arrow back out. Translate each element into a software study example. The source figure is an older author-hosted textbook chapter.

Ask

What could count as an observation for a grid-study agent?

Expected answer

A versioned case file, a timestamped measurement, a solver status or a returned constraint violation.

Transition

Now separate the part that chooses an action from the part that carries it out.

Sources: [DEF01]. Full references follow the glossary.

Open slide 4 online

Slide 5. An agent chooses a step. A tool runs it.

00:05:00-00:06:30 / 1.5 minutes

Say

The agent chooses a permitted step. A tool performs a defined operation and returns a result. For example, the agent can request a power-flow calculation for case F12. The tool interface should identify the case, input versions and settings. The solver then calculates the voltages and flows. The returned evidence should include status and a record that can be inspected. If the solver reports an error, the agent may choose another allowed check or ask for help. The surrounding runtime executes the calls and saves progress. Planning and memory can support that process, although every design need not use separate modules for them. The architecture alone gives no authority to operate equipment.

Show / explain

Point from the core to tools, then back to the returned result. Identify planning and memory as possible supporting roles. Use the run_pf case example rather than interpreting every box as a mandatory component.

Ask

Who calculates the voltage in this example?

Expected answer

The engineering solver. The agent chooses the request and uses the solver's returned evidence.

Transition

A workflow can also call tools, so the next distinction concerns control of the sequence.

Sources: [DEF06], [DEF03], [DEF01], [NV02]. Full references follow the glossary.

Open slide 5 online

Slide 6. Workflow or agent: who chooses the next step?

00:06:30-00:08:00 / 1.5 minutes

Say

In a fixed workflow, the code defines the steps and branches. For example, every case may pass through input validation, power flow and reporting. In an LLM agent, the model chooses some next steps from the allowed actions using the evidence so far. A design can combine both. We may require validation in code while allowing the model to choose which diagnostic tool to call. Anthropic's distinction is useful for engineering discussions, but it does not cover every historical meaning of agent. A loop alone is also insufficient evidence of model-directed choice. Inspect the actual routing rule and the model's role in it.

Show / explain

Follow the model-to-action-to-feedback loop in the documentation figure. Contrast it verbally with a fixed three-step script. Emphasize that a mandatory check can surround a flexible model decision.

Ask

Does a retry loop by itself prove that the model chooses the workflow?

Expected answer

No. Code can implement a fixed retry rule. We need to see how the next action is selected.

Transition

That flexibility makes the task's goals and authority especially important.

Sources: [DEF03], [DEF04], [LG04]. Full references follow the glossary.

Open slide 6 online

Slide 7. An agent needs a clear goal and limits

00:08:00-00:09:30 / 1.5 minutes

Say

A useful assignment states what the agent should achieve, which tools it may use and when it must stop. Our example permits studies and an evidence report. The operator reviews the findings and decides whether a later operating action is allowed. Review may also lead to a request for more analysis. The generic LangGraph form shows one way software can pause for outside input. It is not a utility dispatch screen. The operating process must establish the reviewer's identity, the exact case being reviewed and the permitted response. Approval of an old study should not silently become approval of a changed action or a different grid state.

Show / explain

Read the review form as a pause in a software process. Point out the information a grid-specific review would additionally need: case identity, current versions, evidence and a clearly stated proposed action.

Ask

If a reviewer approves a report, does that automatically authorize live switching?

Expected answer

No. The scope of that approval and the separate operating permissions must be explicit.

Transition

We can now place retrieval, tools, memory and orchestration in the same picture.

Sources: [DEF02], [DEF01], [LG03]. Full references follow the glossary.

Open slide 7 online

Slide 8. A model can use records, tools and memory

00:09:30-00:11:00 / 1.5 minutes

Say

Retrieval finds relevant records. Tools perform defined tasks. Memory preserves context or earlier results. An LLM can use these capabilities while working through a goal. LangGraph can coordinate the execution. PowerAgent develops power-domain models, interfaces and workflows. PowerDAG is a particular research system with its own supervisory design. These names describe different roles, so we should avoid treating them as interchangeable products. The image shows Anthropic's general augmented-model pattern. It is not a diagram of either power-system project. To judge a specific system, ask which component chooses, which component calculates, what evidence is saved and which checks the application enforces.

Show / explain

Identify retrieval, tools and memory around the central model. Give one power example for each: a manual, a power-flow service and a saved study result. Keep the architecture's attribution separate from the later research implementations.

Ask

What is the difference between a retrieved manual and a power-flow result?

Expected answer

The manual supplies documentary evidence. The solver result supplies a calculation for specified inputs. Both need identity and scope checks.

Transition

Let us look at what companies actually report doing.

Sources: [DEF03], [DEF04], [PA02], [PD01]. Full references follow the glossary.

Open slide 8 online

2. What power-sector organizations report

Slide 9. PJM uses HyperQ to review connection applications

00:11:00-00:13:00 / 2 minutes

Say

PJM's interconnection work provides a concrete document-review example. Tapestry's June 2026 report says HyperQ entered Cycle 1 on 29 April, after PJM had confirmed testing in March. The reported Gemini agents check land-use rights, property parcels and tariff evidence for 811 generation applications. They help establish whether application materials are ready for the electrical study. Engineers retain the final decision. This is useful because a study can begin with incomplete or inconsistent evidence. It does not establish autonomous power-flow analysis. The article also discusses a future migration to Google ADK, so we should not label that future runtime as the deployed implementation.

Show / explain

Show the publisher and date in the article header. Identify the pictured animation as publisher material. Connect the document checks to the engineering inputs they precede.

Ask

Which part of an interconnection application is being assisted here?

Expected answer

Document and readiness checks before the electrical study, with final human review.

Transition

The next slide asks exactly what the reported processing numbers count.

Sources: [N01], [N02], [N06]. Full references follow the glossary.

Open slide 9 online

Slide 10. HyperQ: what the reported numbers measure

00:13:00-00:15:00 / 2 minutes

Say

A numerical headline becomes useful only when we know its denominator and timing boundary. Here the supplier reports 811 applications, 4,581 raw documents and 2,328 document bundles. Checking four criteria across those bundles produces 9,312 macro-assessments. That is a count of assessments, rather than a count of projects. The displayed timings cover computer processing and exclude document transfer and human validation. We therefore cannot read the chart as the duration of the entire interconnection process. It also gives no confusion matrix from which to calculate review accuracy. When presenting an efficiency claim, name both the work completed by the system and the work that remains for people.

Show / explain

Read the unit beside each number before discussing speed. Connect 2,328 bundles times four criteria to 9,312 assessments. Point to the timing boundary stated in the caption.

Ask

Can 9,312 assessments be described as 9,312 generation projects?

Expected answer

No. The report covers 811 applications and multiple checks per document bundle.

Transition

An outage-review example uses a different task and a different timing boundary.

Sources: [N01]. Full references follow the glossary.

Open slide 10 online

Slide 11. CAISO/OATI: agents help review planned outages

00:15:00-00:17:30 / 2.5 minutes

Say

The CAISO and OATI example concerns planned-outage review. OATI describes Workflow, Advisor and Recon agents working under a coordinator, with operators retaining authority. CAISO's July 2025 account described a pilot. OATI's later 2026 report describes daily production use and processing more than 500 next-day requests nightly in under eight minutes. That is a batch-processing claim. It implies more than 62.5 requests per minute for the reported batch, but it does not give the latency of an individual request or the remaining reviewer effort. The control-room photograph gives context. It does not independently verify the software or the performance result.

Show / explain

Identify the real control-room photo, then direct attention to the supplier-reported task and batch figures. Distinguish the pilot account from the later production-use statement.

Ask

What measurement would help an operator judge the practical benefit?

Expected answer

End-to-end completion time and human review effort, alongside error rates and the batch processing result.

Transition

Planning-study preparation is another promising place to assist engineers.

Sources: [C01], [C02], [I06]. Full references follow the glossary.

Open slide 11 online

Slide 12. Duke and AWS: agents help prepare planning studies

00:17:30-00:20:00 / 2.5 minutes

Say

AWS reports that Duke engineers use managed agents to prepare cases and coordinate interconnection studies. The September 2026 announcement describes data preparation falling from two weeks of manual work to hours. The agents coordinate existing models, scripts and physics software. Engineers retain responsibility for the engineering decisions. The source does not specify the final number of hours, the sample count or the model and software framework. We can describe the reported change without inventing a precise speedup. Also keep the task clear: faster input preparation is one part of a longer engineering and regulatory process. The operator photograph is contextual and predates the announcement.

Show / explain

Explain that the photograph shows a Duke operator, while the reported application supports planning. On the text, emphasize data preparation as the measured boundary.

Ask

Can we turn two weeks to hours into an exact percentage saving?

Expected answer

No. The final number of hours and the comparison conditions are unspecified.

Transition

Product announcements show possible integration paths, with a different level of evidence.

Sources: [U02], [U03], [I03]. Full references follow the glossary.

Open slide 12 online

Slide 13. Siemens announces agents for PSS E planning studies

00:20:00-00:22:00 / 2 minutes

Say

Siemens announces agentic planning features within Gridscale X and PSS E. The proposed agents coordinate transmission-planning simulations through software interfaces, including connection studies for data centers and other large loads. An established simulation environment offers a practical place to connect an assistant to numerical tools. The announcement does not identify a production user of those new agentic PSS E features. The same release mentions Alliander's migration to the platform, which establishes a different fact. A platform customer is not automatically a user of every newly announced capability. We should keep the feature announcement, customer adoption and measured engineering outcome separate when teaching this case.

Show / explain

Point to the original press-release header and label its evidence type as an announcement. Describe the simulation coordination concept without treating the screenshot as a live application record.

Ask

What additional evidence would establish production use of this feature?

Expected answer

A named user and a clear account of the actual agentic feature in operation, ideally with measured outcomes.

Transition

A direct corporate disclosure gives us another type of use claim.

Sources: [N03]. Full references follow the glossary.

Open slide 13 online

Slide 14. National Grid reports agents checking project plans

00:22:00-00:24:00 / 2 minutes

Say

National Grid's May 2026 results transcript reports agentic checks within its capital control tower. The task is project assurance: examining early project options and documents at approval stages. Feedback arrives while project managers can still revise their plans. This is a useful example of assistance around a decision process rather than direct grid control. The transcript provides no agent framework, accuracy estimate or timing measurement. Later comments about optimizing access outages concern future work. Nearby results about other automation should not be reassigned to this agentic system. Read each claim with its own task, timing and evidence.

Show / explain

Use the slide's three statements as a sequence: project material, agent feedback, manager revision. There is no software screenshot here, so keep the explanation concrete through an example project document.

Ask

What can we conclude from the disclosure alone?

Expected answer

The group reports using agents for project assurance. The source does not quantify their performance.

Transition

Good project evidence also depends on good underlying data.

Sources: [N04]. Full references follow the glossary.

Open slide 14 online

Slide 15. NYPA: better data for agent use

00:24:00-00:25:30 / 1.5 minutes

Say

NYPA's webinar example focuses on data quality. It describes Databricks and Ataccama checks for meter data before AI uses it. Meter records need consistent identifiers, timestamps and quality information to support later analysis. A separate demonstration links Genie to Ataccama through MCP, the Model Context Protocol. That demonstration uses a product and reorder example. We should keep the NYPA data-quality implementation distinct from the generic agent demo. Together they illustrate why agents need reliable inputs and a clear connection to data services. They do not establish a production NYPA agent issuing grid commands.

Show / explain

Identify the NYPA speaker in the original webinar artwork. Explain the separation between the meter-data discussion and the product-data demonstration.

Ask

Why is data quality relevant even if no agent controls equipment?

Expected answer

A planning or advisory answer can still be wrong when it uses missing, duplicated or incorrectly identified data.

Transition

A hiring requirement can show growing interest, while supporting a narrower conclusion.

Sources: [N09], [N10]. Full references follow the glossary.

Open slide 15 online

Slide 16. Eversource: adding agentic skills to grid studies

00:25:30-00:27:00 / 1.5 minutes

Say

Eversource already uses MATLAB and Synergi Electric for distribution studies. A MathWorks account describes conventional Monte Carlo and parallel numerical simulations for examining many possible conditions. Separately, an October 2026 engineering vacancy asks for agentic and AI-based workflows in MATLAB. This shows the company seeking skills that could connect new workflows to existing tools. It does not show that an agentic platform has been deployed. The simulation timing improvement in the older account belongs to conventional automation. Keeping that distinction lets us discuss the motivation honestly: an established engineering environment can make agent integration plausible, while the job posting indicates capability building.

Show / explain

Show the official vacancy header and the agentic-workflow wording. Explain that the earlier simulation work and the vacancy are separate pieces of evidence.

Ask

Does a job posting establish a deployed agentic system?

Expected answer

No. It supports hiring intent and desired engineering capabilities.

Transition

GE Vernova illustrates the platform context into which such capabilities might fit.

Sources: [N07], [N11]. Full references follow the glossary.

Open slide 16 online

Slide 17. GE Vernova: a platform that can connect grid tools

00:27:00-00:28:30 / 1.5 minutes

Say

GE Vernova's GridOS example concerns a platform for coordinating grid software and information. The public announcement names Alabama Power's use of GridOS for Distribution. Shared software can provide places to connect study tools, saved evidence and operator workflows. An agent integrated into that environment would still need clear interfaces, current case identity, engineering checks and bounded authority. The public statement does not isolate a reliability improvement caused by agents or establish an autonomous LLM controller. This case helps us think about adoption infrastructure. We should teach the difference between having an integrated platform and proving that a particular agent improves an operating task.

Show / explain

Use the control-room image to discuss where evidence could reach an operator. Describe interfaces and review as a teaching integration example rather than as verified details of the product.

Ask

What else would we need before claiming an agent improved reliability?

Expected answer

A defined agent task and a study connecting that task to measured reliability outcomes under stated conditions.

Transition

Now consider the people who would review an assistant's work.

Sources: [C07], [I01]. Full references follow the glossary.

Open slide 17 online

3. Operator tasks and evidence checks

Slide 18. Operators often work against a deadline

00:28:30-00:30:30 / 2 minutes

Say

A PNNL survey of 20 control-room operators reported that, on average, 61 percent of tasks needed completion in under 30 minutes. This is an average share of tasks, not the percentage of operators. The survey had 75 questions, but that does not create 1,500 independent timing observations. The source gives no task denominator or spread behind the 61 percent. Treat it as a small self-report study that motivates attention to deadlines. For an assistant to help, it must fit within the time available for finding, checking and acting on evidence. A fast draft that requires extensive correction may still miss that deadline.

Show / explain

Identify the real control-room context. Read the sample size, percentage and time threshold as three separate items. Ask learners to name one time-critical task they actually perform.

Ask

Would a faster model response alone prove a better control-room tool?

Expected answer

No. We must measure the completed task, including review and correction, within its deadline.

Transition

We will use three checks to judge whether a recommendation is useful.

Sources: [Q01], [O01], [I02]. Full references follow the glossary.

Open slide 18 online

Slide 19. Three checks for a useful recommendation

00:30:30-00:32:00 / 1.5 minutes

Say

The first check is data validity. Does the evidence refer to the right asset, time, units and grid connections? The second is engineering validity. Does the study solve the requested problem and meet the relevant equations and limits? The third is decision validity. Does the result justify the next action that this user is allowed to take? Each can fail independently. A well-formatted result can use yesterday's topology. A converged optimization can change a prohibited setting. A correct base-case study can leave required outages untested. These checks give us a consistent way to inspect every later paper and exercise.

Show / explain

Use the conventional situational-awareness screenshot to ask which displayed evidence would need an asset, timestamp and case identity. The three checks are tutorial teaching categories, not measured CAISO agent results.

Ask

If a calculation converges on the wrong topology, which checks are affected?

Expected answer

The input consistency check fails, and the engineering conclusion may no longer apply to the requested grid.

Transition

We begin by turning a broad request into a testable assignment.

Sources: [O05]. Full references follow the glossary.

Open slide 19 online

Slide 20. A clear task makes success easier to test

00:32:00-00:33:30 / 1.5 minutes

Say

Suppose the user asks whether an earlier study still applies after an event. We first identify the relevant asset, event and prior result. A snapshot records the data time, topology version and model revision. The assignment should say what evidence is required and which studies are allowed. It should also specify stopping conditions: missing data, changed inputs, tool failure, exhausted budget or insufficient permission. A successful outcome might be an evidence-backed answer, or a clear request for information needed to finish. The number of tool calls or length of the report does not define success. All F12 labels and policies in this teaching example are invented.

Show / explain

Use the product screen as context for the information involved. Verbally form a request using a named case, snapshot and allowed report. Separate that invented task contract from product capabilities.

Ask

What should the agent return if required current data cannot be obtained?

Expected answer

A clear unresolved status, the missing information and the reason the original question remains unanswered.

Transition

We can now follow an agent's decision loop on that assignment.

Sources: [I09]. Full references follow the glossary.

Open slide 20 online

Slide 21. An LLM agent acts, reads the result and decides again

00:33:30-00:35:30 / 2 minutes

Say

The agent acts, reads the returned evidence and decides again. Imagine that a tool confirms the asset identity but the next tool returns stale measurements. That observation should change the next step. The agent can request a refresh, ask a person or stop with an unresolved result. The surrounding software runs each tool against a specified snapshot and records what happened. A historian returns measurements. A simulator returns modeled behavior. Those outputs answer different questions, even if both arrive as structured data. Feedback in this loop means using returned evidence to choose another step. It does not necessarily mean retraining model weights or learning a new control policy.

Show / explain

Trace the published agent loop from action to tool response to the next decision. Give one normal response and one error response. Ask which path each should trigger.

Ask

When the tool reports stale data, should the original plan continue unchanged?

Expected answer

No. The next step should address the missing current evidence or stop clearly.

Transition

The next example makes that stop condition explicit.

Sources: [DEF03], [LG01], [E01]. Full references follow the glossary.

Open slide 21 online

Slide 22. When data are stale, the next step is to stop

00:35:30-00:37:00 / 1.5 minutes

Say

A successful asset lookup does not guarantee that all later evidence is usable. In this example, the measurement tool marks its result as stale. The system should ask for current evidence and retain the original task as unresolved. It should not replace the missing measurement with an invented value or silently reuse an older one. A trace records the requests, results and case identity so that a reviewer can inspect this behavior. The published picture illustrates a generic gate in a workflow. The gate becomes useful only when the application defines the actual test and enforces it. Saving a trace also does not make the trace correct. It makes inspection possible.

Show / explain

Follow the chain to the gate. State the invented stale-data response and ask learners to choose the next action. Keep the original source figure's generic role clear.

Ask

What should appear in the report after the stale-data check fails?

Expected answer

The missing current evidence, affected case, reason for stopping and question that remains unresolved.

Transition

Even a current result can fail through a unit mismatch.

Sources: [DEF03]. Full references follow the glossary.

Open slide 22 online

Slide 23. Unit checks catch a kV versus p.u. mistake

00:37:00-00:38:30 / 1.5 minutes

Say

Per unit expresses a quantity relative to a chosen base. A measured 12.47 kilovolts equals 1.00 per unit only when the relevant voltage base is also 12.47 kilovolts and the definitions match. If an interface expects per unit, passing the number 12.47 without its unit changes its meaning. The tool should reject that request or require the information needed for an explicit conversion. Software checks should cover the unit, base and asset identity. A prompt saying use correct units is insufficient. After conversion, we still need to check whether the measurement and model are accurate. Unit checking solves one specific consistency problem.

Show / explain

Follow the wrong-value path, identify the missing base, then explain value divided by base. State the matching voltage convention before doing the simple division.

Ask

With a matching 12.47 kV base, what is 12.47 kV in per unit?

Expected answer

1.00 p.u. The conversion still needs the correct asset and voltage definitions.

Transition

The next consistency issue concerns the network itself.

Original teaching explanation or worked example. Preserve its stated assumptions.

Open slide 23 online

Slide 24. A study must match the current grid connections

00:38:30-00:40:00 / 1.5 minutes

Say

The earlier study used topology v42 with a closed switch. The event opens that switch and creates v43. The old result therefore describes a different electrical connection pattern. Updating the report's wording cannot update the voltages or flows. The application must identify the affected calculations, refresh their inputs and run them again. Matching versions is necessary but does not answer every question. We must also ask whether the model accurately represents the real grid and whether the measurements represent the event. This is why evidence should identify both its data snapshot and its model and topology revisions. A recent timestamp alone cannot establish all three.

Show / explain

Point to the changed switch in the before-and-after diagram. Trace which result belongs to v42 and which input snapshot now describes v43.

Ask

Can an old result become valid by changing its topology label?

Expected answer

No. The affected numerical work must actually use the changed network and be checked again.

Transition

We also need to choose the engineering problem that answers the request.

Original teaching explanation or worked example. Preserve its stated assumptions.

Open slide 24 online

Slide 25. A solver result depends on the problem it solves

00:40:00-00:42:00 / 2 minutes

Say

Power flow, optimal dispatch and dynamic simulation answer different questions. Power flow calculates a steady operating condition for specified injections. Optimization chooses permitted variables to improve an objective while meeting stated constraints. A dynamic study follows the system's response over time under a model of equipment behavior. The assistant must select the study that matches the question. A solver's success status belongs to that defined problem. We still inspect numerical residuals, modeled limits, scenario coverage and permissions. A solver can converge perfectly on a problem the user never requested. The generic routing diagram helps us discuss selecting a tool, while the engineering requirements must come from the actual assignment.

Show / explain

Name one question for each study type. Then follow the routing figure and identify the information needed before selecting a branch.

Ask

Does a successful power-flow run establish dynamic stability?

Expected answer

No. It answers a steady-state question. Dynamic stability requires an appropriate dynamic model and study.

Transition

Let us practice checking returned evidence before accepting a conclusion.

Sources: [R01], [L01], [DEF03]. Full references follow the glossary.

Open slide 25 online

Slide 26. Exercise 1: check three returned data packets

00:42:00-00:48:00 / 6 minutes

Say

Work in pairs. For this exercise, evidence must be no more than 60 seconds old, use topology v43 and express the quantity in per unit. The user may recommend actions only. Inspect packets A, B and C, choose which passes these initial rules and explain why the others fail. Then discuss the harder question: does passing these rules establish the complete engineering result or permission to execute anything? The 60-second threshold is an invented classroom rule. Your answer should use the stated contract rather than assumptions about an actual utility's procedures.

Show / explain

Run a six-minute timer: one minute alone, two in pairs, two sharing and one debriefing. Read age_s as age in seconds. Display all three packets throughout.

Ask

Which packet passes, and what remains unproved?

Expected answer

C passes the initial age, version and unit checks. A uses v42. B uses kV. C still needs asset mapping, model and engineering validity, coverage and review. Advisory permission does not authorize execution.

Transition

The same questions help us interpret published research.

Original teaching explanation or worked example. Preserve its stated assumptions.

Open slide 26 online

4. Research foundations, LangGraph and PowerDAG

Slide 27. Power-system papers test different tasks

00:48:00-00:49:30 / 1.5 minutes

Say

The papers ahead investigate several different tasks. Disturbance analysis extracts information from recorded waveforms. Simulation assistants prepare and execute numerical studies. Control prototypes propose changes in model grids. Retrieval benchmarks test whether systems can find and combine evidence correctly. Every result belongs to a particular task, tool set, allowed action space and test population. A percentage from one paper cannot rank a system against another paper that measures something else. As we proceed, use four questions: what does the agent decide, what do the tools calculate, what counts as success and which failures remain? This will make the architectures easier to compare without collapsing their different goals.

Show / explain

Treat the source photograph as research context. Ask learners to place their own task into one of the study families before introducing the first framework.

Ask

Why should we avoid ranking all papers by their reported percentages?

Expected answer

Their tasks, datasets, scoring rules and allowed computation differ. The percentages answer different questions.

Transition

PowerAgent first provides a roadmap for organizing power-domain components.

Sources: [D01], [E02], [E03], [P02], [I04]. Full references follow the glossary.

Open slide 27 online

Slide 28. Harvard PowerAgent: models, tools and study workflows

00:49:30-00:51:30 / 2 minutes

Say

Harvard's PowerAgent roadmap connects three layers. PowerFM concerns models that support power-system tasks. PowerMCP concerns interfaces through which software can expose engineering tools. PowerWF concerns workflows that coordinate the work. The 2025 IEEE article is a roadmap, rather than a benchmark claiming one fixed deployment result. PowerAgentBench-Dyn is a later benchmark with its own task and evaluation rules. Keeping the initiative and the benchmark separate avoids assigning one experiment's score to the whole project. The architecture shown here comes from the authors' presentation. We inspected that presentation and the project materials, while the original article's full experimental content was unavailable. We therefore attribute no uninspected experimental result to it.

Show / explain

Walk through the three source panels in order: models, interfaces, workflows. Give a short example of each role. Cite the IEEE title and DOI in the script's source section.

Ask

Where would an interface to an existing simulator fit in this roadmap?

Expected answer

In the tool-connection role described by PowerMCP, while the workflow controls when and how it is used.

Transition

A concrete PowerWF example makes these layers easier to understand.

Sources: [PA01], [PA03], [PA02], [P01]. Full references follow the glossary.

Open slide 28 online

Slide 29. PowerAgent example: what changes when load is added?

00:51:30-00:53:30 / 2 minutes

Say

Consider a proposed load addition. The workflow analyzes an original case and a case with added load. It runs power flow and selected N minus one checks, which examine single-component outages. LangGraph coordinates the steps and saved results. PowerWorld performs the electrical calculation. That separation tells us what each component contributes: coordination, numerical analysis and evidence for comparing the cases. The maintained graph code uses StateGraph and ToolNode, but one repository does not establish the software choice for every PowerAgent project. The demonstrations provide a concrete workflow. They do not give a controlled estimate of utility time savings or operating reliability. We still need the cases, assumptions and comparison results.

Show / explain

Follow the source workflow from the load-growth request into base and changed-case studies. Point to the numerical comparison rather than treating the picture as an exact rendering of the current code.

Ask

What changes when we add load: the language model or the electrical study inputs?

Expected answer

The electrical inputs change. The workflow must run and compare the appropriate studies for those inputs.

Transition

Next, a disturbance-analysis paper shows why tool order matters.

Sources: [PA06], [PA03], [PA05]. Full references follow the glossary.

Open slide 29 online

Slide 30. DrAgent: reading a recorded grid fault

00:53:30-00:55:30 / 2 minutes

Say

DrAgent studies recorded disturbances on transmission lines. First look at the waveform before discussing any AI component. The third current trace changes sharply during the fault. The marked locations identify inception, the highest current and clearance, when the fault ends. A tool can return a sample index or a current measurement. An engineering answer still needs to translate that output into the requested quantity and units. Locating a visible spike is a different question from interpreting its physical cause or assessing equipment. The figure is an original research record, rather than an invented control-room event. We will next connect the waveform questions to the tools that answer them.

Show / explain

Locate the third trace. Point to the start, peak and clearance markers in that order. Explain that the horizontal position represents time or sample location and the vertical value represents current.

Ask

Would the highest current sample alone tell us the fault duration?

Expected answer

No. Duration needs a start and an end, with the appropriate timing information.

Transition

The architecture assigns those subtasks to particular tools.

Sources: [D01]. Full references follow the glossary.

Open slide 30 online

Slide 31. DrAgent selects tools to answer a fault question

00:55:30-00:57:30 / 2 minutes

Say

DrAgent uses a ReWOO-based plan to choose domain functions and document search. ReWOO plans tool work using references to results that earlier steps will produce. In this implementation, Gemini 1.5 Pro plans, about a dozen domain functions analyze signals, and retrieval searches 138 product manuals. Recorded disturbance files use the COMTRADE format. Notice the two different evidence paths. Signal functions measure the event, while manual search supplies product information. A question about a fault's start time may need its sample index and the recording's sampling rate before conversion to seconds. That requirement is a dependency between calculations, which gives us a reason to introduce task graphs.

Show / explain

Follow the planning stage into signal analysis and manual search, then back into the response. Explain RAG as finding relevant source passages for the model.

Ask

Which two inputs are needed to turn a sample index into a time offset?

Expected answer

The index under the tool's indexing convention and the sampling rate. A record start time is also needed for an absolute timestamp.

Transition

Before graph design, inspect what this tool-based approach achieved.

Sources: [D01], [LG01]. Full references follow the glossary.

Open slide 31 online

Slide 32. DrAgent: more correct answers, with errors still present

00:57:30-00:59:30 / 2 minutes

Say

The chart reconstructs correct-answer counts from the paper's individual score cells. Across 18 recorded cases, fault-inception answers increase from 12 to 18, and maximum-current answers from 16 to 18. The timing category increases from four to 13. Here we count scores of two or above as correct under the authors' rubric. The timing category also accepts a correctly labeled duration, so it is broader than exact clearance timestamps. A domain expert graded the answers using the paper's tolerances of three milliseconds for timing and five percent for current. These are author-data-derived counts, rather than a new experiment. They show useful gains on this task while leaving errors and a small test population.

Show / explain

Read the denominator before the bars. Compare the timing bars last, then identify the remaining five cases. Mention the audited counts rather than the inconsistent printed Q6 mean.

Ask

Does 18 out of 18 establish general fault-analysis reliability?

Expected answer

No. It describes the tested records and grading rule. Other records and tasks remain to be evaluated.

Transition

The tool sequence becomes clearer once we distinguish three kinds of graph.

Sources: [D01]. Full references follow the glossary.

Open slide 32 online

Slide 33. Three kinds of graph have different jobs

00:59:30-01:01:30 / 2 minutes

Say

The first graph is the electrical network. Its edges represent physical connections between modeled components. The second is a task dependency graph. Its edges say which calculation needs an earlier result. The third is an execution workflow. Its edges determine which software step can run next, and it may contain retry loops. These are different objects. A radial feeder can have a tree-like electrical shape while its software repeatedly revisits a diagnostic step. A calculation DAG can analyze a meshed electrical network. When someone says a node or a loop, ask which graph they mean. This prevents a software architecture picture from being mistaken for the grid model.

Show / explain

Compare the three panels. Name one node and the meaning of one edge in each. Point out the loop in the workflow and its absence from the task DAG.

Ask

Does a loop in the software imply that the electrical feeder is meshed?

Expected answer

No. Software execution and electrical connectivity are separate structures.

Transition

Now use the fault example to read an actual dependency graph.

Sources: [D01], [P03], [LG01]. Full references follow the glossary.

Open slide 33 online

Slide 34. A DAG shows which result must come first

01:01:30-01:03:30 / 2 minutes

Say

A directed acyclic graph, or DAG, connects tasks without forming a directed loop. An arrow means the downstream calculation needs the upstream result. To convert a fault-start sample into a time, we first need the sample index and the sampling rate. A peak-current calculation may run independently when it uses the same unchanged record and does not need the timing result. A valid execution order respects every required dependency. The page position of a box does not determine its order. Before reporting, the application must wait for all required outputs or clearly identify what is missing. This is a practical way to reason about both correctness and possible parallel work.

Show / explain

Trace both arrows into the time conversion. Then follow the separate peak-current branch. Ask learners to identify a valid order without relying on left-to-right box placement.

Ask

Can the time conversion run as soon as the sample index arrives?

Expected answer

Only if the required sampling-rate information is already available and matches the same record.

Transition

LangGraph manages execution, which can include more than an acyclic task plan.

Sources: [D01]. Full references follow the glossary.

Open slide 34 online

Slide 35. LangGraph can pause to inspect a run

01:03:30-01:05:30 / 2 minutes

Say

LangGraph is software for defining state and running workflow nodes with edges and routing rules. A node can call fixed code, a simulator or an LLM. Unlike a DAG, the execution can revisit a step. A debugger can pause the graph so a developer can inspect progress. That static breakpoint differs from a dynamic interrupt that requests external input. StateGraph defines stored fields, while application reducers determine how parallel updates combine. Those facilities help implement a process. The application still defines data validation, engineering tests, permissions and stopping conditions. Executing the graph successfully cannot establish that the electrical model or the chosen study is valid.

Show / explain

Explain the highlighted debug pause as a developer inspection point. Use a short run-check-repair example to describe a loop, then distinguish it from human review.

Ask

Does using LangGraph automatically add a power-system feasibility check?

Expected answer

No. The application must call and interpret the appropriate engineering tools and enforce the requirements.

Transition

Those tools produce evidence that the workflow must keep identifiable.

Sources: [LG01], [D01], [P03], [LG03]. Full references follow the glossary.

Open slide 35 online

Slide 36. Saved state must identify the study and its evidence

01:05:30-01:07:30 / 2 minutes

Say

Saved state should identify the study, its input versions, results, status and attempt count. Suppose a set of parallel outage studies expects a case called outage_34. The application needs to notice if that result is missing or appears twice. A generic state store does not automatically understand that engineering requirement. Reducers or an external evidence store must preserve the case identities and combine updates correctly. A list of three completed steps is insufficient if the requested set contained four cases. Save enough information to compare the expected set with the returned evidence. That makes both missing coverage and inappropriate reuse easier to detect.

Show / explain

Read the figure as generic software state. Describe expected-case and returned-case sets verbally. Make clear that those grid-specific fields are tutorial design requirements.

Ask

What should the report do when one required outage result is absent?

Expected answer

Name the missing case and retain an incomplete or unresolved status for conclusions that depend on it.

Transition

Saved progress also allows a run to pause and resume.

Sources: [LG01], [LG02], [LG05]. Full references follow the glossary.

Open slide 36 online

Slide 37. A checkpoint lets a workflow pause and resume

01:07:30-01:09:30 / 2 minutes

Say

A checkpoint records workflow progress. A thread identifier links saved steps to the same run so execution can resume later. However, external conditions may change during a pause. A saved voltage result may no longer match the current topology. The application must revalidate the snapshot and rerun affected work. There is another software detail to consider: the interrupted node may restart from its beginning. Actions before the pause may therefore run again. Stable step IDs or idempotency controls can prevent a repeated request from applying the same change twice. Saving and resuming software state solves a coordination problem. It does not keep the external grid state frozen.

Show / explain

Follow the checkpoint sequence across the pause. Ask what could change outside the software before resumption. Explain idempotency as safely recognizing a repeated operation.

Ask

Can resuming a saved thread justify reusing every old numerical result?

Expected answer

No. Reuse depends on whether each result's relevant inputs still match.

Transition

The same identity checks matter when several studies run together.

Sources: [LG02], [LG03], [LG05]. Full references follow the glossary.

Open slide 37 online

Slide 38. Parallel outage studies must account for every case

01:09:30-01:11:30 / 2 minutes

Say

Parallel studies can reduce waiting when the cases are independent. Each selected outage should run on a consistent copy of the same starting case. Afterward, the application must compare the expected case IDs with the returned results. A failed or missing branch remains unresolved. An aggregator that summarizes only successful branches can produce an attractive but incomplete report. If a case called C2 fails, the report should identify C2, its error and the conclusions that cannot yet be drawn. Even completing every selected case does not imply that the selected set covers every relevant single-component outage. Coverage has to match the original assignment.

Show / explain

Trace the generic fan-out and fan-in figure. Verbally remove one branch result and ask what the aggregator must preserve rather than summarize away.

Ask

If two out of three required studies finish, can the report declare all selected outages acceptable?

Expected answer

No. It can report those two results while explicitly leaving the third case and affected conclusions unresolved.

Transition

PowerDAG addresses a related problem: calling a tool before its prerequisites exist.

Sources: [R01], [LG01], [DEF03]. Full references follow the glossary.

Open slide 38 online

Slide 39. PowerDAG checks the steps needed before a tool runs

01:11:30-01:13:30 / 2 minutes

Say

PowerDAG retrieves relevant verified workflow examples and uses a dependency graph with a supervisory gate. The gate checks required earlier tool calls before allowing another tool to run. The inspected version four covers 108 tools, ten task families and four Vermont feeder models. Those are distribution networks: Rochester, Stowe, Glover and South Hero. The agent still chooses steps, while the supervisor constrains their order. This combines flexible selection with explicit workflow rules. The exact gate matters. In this paper it checks encoded prior Write-tool names. That is narrower than checking every parameter, input version or physical consequence. We should explain the gate's actual guarantee before interpreting its benchmark result.

Show / explain

Follow the agent's proposed call through the supervisor and into the environment. Show where retrieval supplies examples and where the prerequisite check occurs.

Ask

Can a prerequisite-name gate alone prove that a result uses the current topology?

Expected answer

No. Freshness and parameter consistency need additional checks that identify the actual inputs and outputs.

Transition

The voltage example makes that boundary concrete.

Sources: [PD01], [PD02]. Full references follow the glossary.

Open slide 39 online

Slide 40. PowerDAG: solve the grid before checking its voltage

01:13:30-01:15:30 / 2 minutes

Say

A voltage-violation check needs a solved network state. If the required earlier solver step is missing, the gate rejects the call and explains what needs to happen first. That prevents one important class of workflow error. Now ask a second question: did that solve use the current network? The paper's check of executed tool names does not by itself establish freshness. Our teaching diagram adds a current-v43 condition to show the stronger requirement an application may need. It is labeled as an adaptation. Passing a dependency gate therefore means the encoded prerequisites passed. The numerical solution and the resulting engineering conclusion still require their own checks.

Show / explain

Read the missing-prerequisite path first, then the permitted path. Point explicitly to the added v43 freshness condition so learners do not attribute it to the paper's original gate.

Ask

If solve appears in the history, what else would you want before checking voltage?

Expected answer

Evidence that the solve used the required case and current inputs, returned an appropriate status and produced the needed results.

Transition

Now interpret the controlled benchmark without enlarging that guarantee.

Sources: [PD01]. Full references follow the glossary.

Open slide 40 online

Slide 41. PowerDAG results: the same queries, four setups

01:15:30-01:17:30 / 2 minutes

Say

This comparison uses the same GPT-5.5 model and 150 test queries in each setup. The reported first-run workflow success is 77.33 percent for ReAct, 92.67 for retrieval only, 96 for just-in-time supervision only and 98 for full PowerDAG. Pass at one checks required tool coverage, compatible inputs and ordering against expert workflows. Each query runs once in each setup. These held-out queries still come from the same ten task families. The comparison supports the value of these complete configurations on that benchmark. It does not measure reliability of live commands or isolate the effect of one graph edge. Keep the model, task population and score definition attached to the headline.

Show / explain

Read the same-model label and denominator before comparing bars. Explain retrieval only and supervision only as separate tested setups. Finish by defining what the 98 percent actually measures.

Ask

Does 98 percent workflow Pass@1 mean a 98 percent chance of safe field operation?

Expected answer

No. It measures the specified first-run workflow criteria on the benchmark queries.

Transition

Reproducibility also depends on what the public release contains.

Sources: [PD01]. Full references follow the glossary.

Open slide 41 online

Slide 42. PowerDAG: what can others check and reuse?

01:17:30-01:19:30 / 2 minutes

Say

The PowerDAG public benchmark provides tasks, tool descriptions, workflow records and scoring code. That allows others to inspect the study design and some of the execution evidence. The release has a practical boundary: BESS, basicPV and GFI can run from the public files, while seven other task families need private feeder maps, geographic data and network-tool code. Public records are therefore useful without making the entire study independently runnable from the download. When reusing the idea, we can adopt explicit prerequisite checks and then test them against our own tools and state rules. We should not assume that a published gate automatically transfers unchanged to a different utility environment.

Show / explain

Count the three public task families and the seven with external dependencies verbally. Point learners to the linked paper and repository rather than promising a one-click full reproduction.

Ask

What can we reuse even when every dataset is unavailable?

Expected answer

The design idea, task and tool descriptions, available workflow evidence and scoring approach, followed by new validation for the intended environment.

Transition

Another paper focuses on generating and repairing simulation code.

Sources: [PD02], [PD01]. Full references follow the glossary.

Open slide 42 online

5. Simulation feedback and valid reuse

Slide 43. Simulation agents find functions, run code and repair errors

01:19:30-01:21:30 / 2 minutes

Say

The feedback-driven simulation framework finds relevant MATLAB functions and settings, writes code and runs it. Tool errors return feedback that can guide repair. The study uses DALINE and MATPOWER tasks. This illustrates a real agentic role: selecting and correcting the steps needed to complete a simulation request. However, a program can run without matching the requested engineering settings. We must compare the executed study with the assignment, including required variables and unnecessary changes. A successful software exit is one layer of evidence. Numerical and task-level checks are additional layers. The paper does not identify LangGraph, so we should describe its workflow without assigning an unreported software framework.

Show / explain

Follow the source feedback path from generated code to execution and back to repair. Give an example of code running with the wrong setting and ask where that error would be detected.

Ask

Is no MATLAB error a sufficient success criterion?

Expected answer

No. The study and settings must also satisfy the requested engineering task.

Transition

The scoring rule in this paper makes that distinction particularly important.

Sources: [E02]. Full references follow the glossary.

Open slide 43 online

Slide 44. Simulation-agent results: points, not a pass rate

01:21:30-01:23:30 / 2 minutes

Say

For 34 DALINE tasks, each graded attempt receives 100 points if it meets every requirement without extra settings, 50 if unnecessary settings remain and zero if a requirement is missed. The score averages allowed attempt slots, carrying the final score into unused slots. The full setup reports 93.13 on this scale. It is not a statement that 93.13 percent of tasks passed. Other shown setups score zero, 31.37 and 33.82. These compare complete systems with different retrieval and feedback arrangements, including a separate ChatGPT product setup. The related MATPOWER test reports 96.85 over 35 tasks with up to five attempts. Always explain the score before comparing its height.

Show / explain

Read the horizontal-axis metric and the 100/50/0 rubric aloud. Compare the full setup with the others only after explaining the attempt averaging.

Ask

Can we multiply 34 tasks by 93.13 percent to recover a number of successful tasks?

Expected answer

No. The graded attempt-averaged score is not a completion fraction.

Transition

Repeated feedback can help, but it can also preserve a software mistake.

Sources: [E02]. Full references follow the glossary.

Open slide 44 online

Slide 45. Trying again can repeat the same software error

01:23:30-01:26:00 / 2.5 minutes

Say

A capability study found an error involving a 100 megawatt and 10 megavar load request. The agent request omitted reactive load, and the cache key omitted it as well. Correcting the request still returned the old result because the cache could not distinguish the corrected input. A cache stores earlier results for reuse. Its key must include the inputs that determine whether reuse is valid. In this case, trying again with better wording cannot repair the underlying key. The lesson extends to model revisions, topology and other relevant settings. Agent evaluation needs to test adapters and caching behavior as well as model responses. A fluent explanation can repeatedly describe the same stale calculation.

Show / explain

Use the evaluator loop as a generic feedback picture. Narrate the original request, corrected request and unchanged cache result. Ask which software component must change.

Ask

Why did correcting the prompt fail to obtain a fresh result?

Expected answer

The cache key ignored a result-changing input. The adapter and cache needed correction, not another verbal retry.

Transition

Let us make these loops visible in two official platform demonstrations.

Sources: [R02], [DEF03]. Full references follow the glossary.

Open slide 45 online

Slide 46. LangGraph Studio: watch an agent use tools

01:26:00-01:30:00 / 4 minutes

Say

This official walkthrough makes an agent run visible. Watch the graph, the selected tool and the returned result. The platform can pause a run and inspect its saved state. In a power-system application, the numerical solver would be one of these tools. The model coordinates the request; the solver computes the electrical result; the operator decides what to do. This is the original 2024 Studio interface, not evidence of a utility deployment.

Show / explain

Click Play official demo. Follow one visible request through the agent graph. Pause when a tool result or delegation event is shown. Use the direct watch link if presenting in a separate tab. Ask learners to name the tool, result and next decision; do not treat the demo as grid validation.

Ask

Where would you pause this workflow to check a power-flow result?

Expected answer

After the solver returns and before a recommendation is accepted. Check convergence, units and the study inputs.

Transition

Now compare how work and evidence move through the next example.

Sources: [VID01]. Full references follow the glossary.

Open slide 46 online

Slide 47. Siemens PSS E: agentic transmission planning

01:30:00-01:34:00 / 4 minutes

Say

Play the full 79-second product video linked from Siemens' agentic transmission-planning page. This is a supplier overview of Gridscale X PSS E, rather than a complete recorded engineering study. Connect its planning message to three separate jobs: preparing a study, computing the electrical result and interpreting it. The solver remains essential. In our lab we inspect those steps explicitly. A product video alone does not establish field accuracy or deployment at a named utility.

Show / explain

Play the full 79-second Siemens product video. Then connect its planning message to study preparation, the PSS E solver and engineering review. Ask learners which inputs and numerical checks are needed. The overview does not display a complete tool-call trace.

Ask

Which part must still be checked by a power engineer?

Expected answer

The study inputs, the solver's convergence and limits, and whether the result supports the proposed decision.

Transition

Return to power-system studies: keep the same question about who supplies and checks each result.

Sources: [VID02]. Full references follow the glossary.

Open slide 47 online

Slide 48. GridMind 2025 connects typed agents to solvers

01:34:00-01:36:00 / 2 minutes

Say

The 2025 GridMind study uses PydanticAI to structure agents and pandapower for AC OPF and outage studies. It tests 14-, 30-, 118- and 300-bus systems. On case 118, the paper reports 100 percent ACOPF success across six models, with timing distributions based on five repeats. Structured types help make inputs and outputs machine-readable, but they do not establish that every required outage was studied or interpreted correctly. Keep that task boundary attached to the result. Also keep this 2025 system separate from the 2026 weather-risk GridMind workflow that we discuss next. They share a name while using different architectures and addressing different tasks.

Show / explain

Trace the typed-agent architecture into the solver. State the year and the ACOPF task explicitly. Do not infer an undisclosed success denominator from the timing sample.

Ask

Does valid structured output prove complete contingency analysis?

Expected answer

No. We need the requested case list, returned evidence and an independent coverage check.

Transition

The paper's reported answers show why agreement alone is insufficient.

Sources: [P04]. Full references follow the glossary.

Open slide 48 online

Slide 49. GridMind: reported answers need independent checks

01:36:00-01:38:00 / 2 minutes

Say

For case 118, one model reports 165 percent maximum overload and five report 137 percent. The models also identify different critical lines. These are different model setups, not six repeats that estimate one model's error probability. The table provides no complete independent outage reference that would tell us which answer covers the full required scope. A majority answer may be shared, incomplete or based on different selections. The response screenshot itself is an example of agent output, rather than ground truth. To resolve the discrepancy, obtain the same case definitions and run or inspect an independent reference study. Count agreement only after establishing what the answers were supposed to cover.

Show / explain

Use the response screenshot to distinguish a reported claim from its supporting evidence. Compare the overload figures verbally and ask what reference is missing.

Ask

Can five agreeing models establish that 137 percent is correct?

Expected answer

No. Agreement does not replace independent evidence for the same defined study scope.

Transition

The later GridMind study concentrates on state reuse across a weather workflow.

Sources: [P04]. Full references follow the glossary.

Open slide 49 online

Slide 50. GridMind 2026 reuses results when their inputs still match

01:38:00-01:40:30 / 2.5 minutes

Say

The 2026 GridMind workflow coordinates five stages using LangGraph: weather, outages, ACOPF, screening and cascade analysis. Weather comes from AERIS. Base OPF uses LUMINA or pandapower. A LUMINA surrogate provides faster screening, and EPFast uses a reduced-order DC model for cascade simulation. Input fingerprints identify when a saved result can be reused. A fingerprint is a hash of declared inputs. It can show whether those inputs match an earlier run, but it cannot detect an important input that was never included. Dependencies must reflect the actual implementation. Here the contingency stage reads outage schedules directly, so a change in the ACOPF solver does not automatically invalidate every stage that appears later in a list.

Show / explain

Walk through the five named stages verbally. For each, name its output and which later work actually consumes that output. Distinguish a surrogate from a detailed physics calculation.

Ask

What must a fingerprint include to support safe reuse of a numerical result?

Expected answer

Every declared input that can change that result, including the relevant model, settings and external data versions.

Transition

Let us use specific input changes to read those dependencies.

Sources: [P03], [P03A]. Full references follow the glossary.

Open slide 50 online

Slide 51. GridMind: which changes need a new run?

01:40:30-01:42:30 / 2 minutes

Say

If only the cascade parameter k changes, this implementation reruns cascade and reuses the unaffected results. If the forecast initialization shifts by six hours, all five stages rerun. A change of ACOPF solver reruns ACOPF only in this particular implementation because its other stages do not all consume that output. These examples show why a workflow's dependency structure matters more than a simple left-to-right sequence. Returning to exactly matching inputs can permit reuse of an earlier result. That reuse still depends on the key recording every relevant input. Another workflow may have different dependencies, so these rules should travel with the implementation rather than become universal weather-study rules.

Show / explain

Trace the highlighted invalidation paths for each change. Explain why an arrow represents actual data dependence, rather than visual proximity.

Ask

If only k changes, should all five stages rerun under the cited design?

Expected answer

No. Cascade reruns, while results whose relevant inputs still match may be retained.

Transition

Reusing fewer stages can save time, but the two percentages need separate measurements.

Sources: [P03]. Full references follow the glossary.

Open slide 51 online

Slide 52. GridMind results: fewer stages and less time are different

01:42:30-01:45:00 / 2.5 minutes

Say

For the same seven-query session, the paper reports a 60 percent reduction in stage runs and about a 52.7 percent reduction in total run time. Those quantities differ because stages have different costs. Avoid calling a 60 percent reduction in stage count a 60 percent time saving. The gain comes from workflow design and appropriate reuse, rather than simply from a faster language model. The study reports one seven-query session and gives no repeated-session interval for the time reduction. It also uses surrogate and reduced-order models. We can learn a useful design principle while retaining those evaluation limits and the need for engineering validation.

Show / explain

Read stage executions and elapsed time as separate panels or quantities. Ask which stages dominate time and what additional measurements would establish variability.

Ask

Why need the time saving differ from the reduction in stage count?

Expected answer

The omitted and retained stages can take different amounts of time, and execution overhead also contributes.

Transition

We will practice choosing which saved results still apply.

Sources: [P03]. Full references follow the glossary.

Open slide 52 online

Slide 53. Exercise 2: check whether saved weather-risk results still apply

01:45:00-01:53:00 / 8 minutes

Say

Start from a saved run with forecast F0, topology v43 and cascade parameter k equal to 50. First, change only k to 200 and decide which stages need new results. Next, shift the forecast initialization by six hours. Use the dependencies we just discussed, rather than assuming every implementation behaves the same way. Finally, imagine a required simulation fails. Decide what the operator should receive and which conclusion remains unsupported. Work from the specific cited workflow. F0 and v43 are invented classroom labels. The goal is to identify the valid evidence set and communicate any gap clearly.

Show / explain

Run an eight-minute timer: two minutes alone, two comparing in groups, two discussing and two debriefing. Keep the five stages visible while groups answer.

Ask

What reruns, and how should a failed required stage be reported?

Expected answer

Only k changed: rerun cascade. Forecast shifted: rerun all five. A failure requires incomplete or unresolved status with case and stage IDs, input versions, error and missing conclusions.

Transition

A dynamic-model benchmark makes the allowed action space equally explicit.

Sources: [P03], [LG02]. Full references follow the glossary.

Open slide 53 online

6. Power-system benchmarks and failure modes

Slide 54. PowerAgentBench-Dyn allows only specific model changes

01:53:00-01:55:30 / 2.5 minutes

Say

PowerAgentBench-Dyn is a Harvard-led benchmark, separate from the broader PowerAgent roadmap. In its model-quality task, an agent receives a modified solar-PV dynamic model that fails eight quality tests. DMView and PSS/E provide feedback. The agent may change only four REECAU1 controller gains, with all other fields locked and at most five parameter updates. A gain changes a controller's response within the simulation model. The benchmark asks whether a tool-using agent can repair that specified model under clear constraints. It does not authorize changing a utility's live controller. The public harness also needs licensed PSS/E and external DMView software, which matters for audience expectations about reproduction.

Show / explain

Follow the source benchmark cycle from permitted parameter edits to tests and feedback. Identify the locked fields and update budget as part of the task definition.

Ask

Why is the allowed four-gain set part of the success criterion?

Expected answer

A solution obtained by changing prohibited fields would solve a different task, even if the test outputs improved.

Transition

The results distinguish completing the test suite from the amount of search needed.

Sources: [P01], [P01C]. Full references follow the glossary.

Open slide 54 online

Slide 55. PowerAgentBench-Dyn results: passed sessions and updates

01:55:30-01:58:00 / 2.5 minutes

Say

Each session passes only when all eight quality tests pass within five allowed updates. On the one modified solar-PV model, the paper reports 10 successful sessions out of 10 for Opus 4.8, 10 out of 10 for Sonnet 4.6 and nine out of 10 for Haiku 4.5. Median updates are one, one and two. The left-hand counts describe complete compliant sessions, while the right-hand values describe update effort. They are different measures. Testing one modified model limits generalization, and the study gives no matched human or fixed-search baseline. These results therefore do not establish a comparative speedup or fleet-wide model-repair reliability.

Show / explain

Read the success counts first, including the single failed session. Then read the median-update panel and explain that it does not include all operator work.

Ask

Does a median of one update tell us the total human time saved?

Expected answer

No. It describes model-update effort in this test, not an end-to-end comparison with human work.

Transition

The same benchmark also includes a distinct dynamic-security task.

Sources: [P01]. Full references follow the glossary.

Open slide 55 online

Slide 56. PowerAgentBench-Dyn also checks grid response to faults

01:58:00-02:00:30 / 2.5 minutes

Say

The dynamic-security task uses PowerMCP and PowerFactory to study ten candidate faults and identify three critical cases. Full-workflow success also requires correct findings and staying within the allowed synchronous-condenser mitigation options. Across six models, reported passes range from zero out of ten to ten out of ten. Opus 4.7 and Sonnet 4.6 each pass nine, Haiku 4.5 passes seven, Gemini 3.1 Pro and GPT-5.5 each pass ten, and Qwen 2.5-Coder-7B passes zero. This task uses Opus 4.7, which differs from the model-quality task's Opus 4.8. A run can fail at tool execution before reaching an engineering finding. Keep those failure stages separate.

Show / explain

Explain the study as fault execution, identification and constrained mitigation. Read complete-workflow counts without combining them with the previous model-repair sessions.

Ask

Can these counts be combined with the previous slide into one general reliability rate?

Expected answer

No. The tasks, model versions and pass criteria differ.

Transition

Now compare a different tool-based power-system assistant, VeraGrid-Agent.

Sources: [P01]. Full references follow the glossary.

Open slide 56 online

Slide 57. VeraGrid-Agent uses solver results to answer a question

02:00:30-02:02:30 / 2 minutes

Say

VeraGrid-Agent writes feeder inputs, runs AC optimal power flow in VeraGrid and reads the relevant output rows. Within a fixed budget it can repair tool errors, then return a multiple-choice answer. The benchmark uses 33 templates and 150 questions on one balanced radial eight-bus feeder, with 50 questions at each of three difficulty levels. Reference answers come from the same solver and circuit. The task tests whether the complete tool-using setup answers those questions accurately. It does not establish a general feeder-control policy. The same numerical study can succeed while the agent reads the wrong row, so solving and interpreting remain separate failure points.

Show / explain

Trace input writing, solving, reading and final answering in the source workflow. Identify the one-feeder and multiple-choice boundaries.

Ask

If the solver succeeds, can the final answer still be wrong?

Expected answer

Yes. The agent may read or interpret the wrong result, use the wrong quantity or return the wrong answer choice.

Transition

The comparison shows the value of that whole tool-assisted setup on its defined questions.

Sources: [E04], [E04C]. Full references follow the glossary.

Open slide 57 online

Slide 58. VeraGrid-Agent results: the same models with and without tools

02:02:30-02:04:30 / 2 minutes

Say

The chart compares the same named models with and without the solver-agent arrangement on 150 questions. GPT-5.2 rises from 64 correct answers to 149. Claude Sonnet 4.5 rises from 71 to 149, and Gemini 3 Flash from 74 to 150. The no-tool setup receives feeder and distributed-resource information, while the agent adds solver execution, reading tools and repeated feedback. This is a comparison of complete systems with different computation, rather than an isolated estimate of one feature's effect. The questions all use one feeder, and repeated-run variation is unspecified. The remaining errors and task boundary matter even when a bar is close to the maximum.

Show / explain

Read every pair with its denominator, then point to the one-feeder caption. Ask what additional tests would be needed for another feeder or operating task.

Ask

Does 150 out of 150 establish general feeder dispatch performance?

Expected answer

No. It establishes the reported answer count on this fixed benchmark, with its solver, circuit and question templates.

Transition

The next study shows why staying within the assigned decision space matters.

Sources: [E04]. Full references follow the glossary.

Open slide 58 online

Slide 59. AgentiGrid: what does one reported run mean?

02:04:30-02:06:30 / 2 minutes

Say

AgentiGrid connects an agent to ExaGO for numerical grid-planning studies. The paper tests four models on three applications: dispatch exploration, outage screening and studies with uncertain inputs. Those combinations produce 12 different searches on one 200-bus model. They are not 12 repeats of the same task. A run without a fatal JSON error has shown that an interface exchange worked, but the complete task can still fail. We must check allowed changes, coverage of the requested cases and faithful interpretation of solver evidence. This paper is useful because its reported examples expose those different failure modes within an otherwise functioning software process.

Show / explain

Follow the prototype architecture into ExaGO. State four models times three applications as the source of the 12 setups, then separate interface execution from engineering completion.

Ask

Can 12 different searches estimate the repeatability of one fixed task and setup?

Expected answer

No. They describe different configurations and applications.

Transition

One search found a lower objective by changing the problem itself.

Sources: [R01]. Full references follow the glossary.

Open slide 59 online

Slide 60. A lower cost can come from solving the wrong task

02:06:30-02:08:30 / 2 minutes

Say

The reported objective falls from 27,564 to 17,398, a reduction of about 36.9 percent in unspecified objective units. That looks attractive until we inspect the allowed changes. The task permitted dispatch changes, meaning the outputs of units already selected to operate. The model changed unit commitment, meaning which units are on. The equation shows why a larger decision set can have an equal or lower minimum. It gives the optimizer more choices. That lower value therefore does not prove improvement on the original dispatch-only task. First verify the permitted action space, then compare the objective under the same constraints and study conditions.

Show / explain

Read the subset relation as one allowed decision set being contained in a larger set. Explain the minimum inequality in words, then connect it to dispatch versus commitment.

Ask

Why can a numerically lower objective still fail the task?

Expected answer

It may use prohibited decisions or a different constraint set, so it no longer answers the assigned optimization problem.

Transition

Incomplete outage coverage creates another task-level failure.

Sources: [R01]. Full references follow the glossary.

Open slide 60 online

Slide 61. Successful tool calls can still miss a critical outage

02:08:30-02:11:00 / 2.5 minutes

Say

All four reported outage searches find the highest-cost outage, involving lines 187 to 189. Three find the infeasible outage 29 to 30. None finds the second infeasible outage, 15 to 16. Here infeasible means that the specified study has no solution meeting its constraints. Finding the expensive outage does not establish that the search found every critical case. These are four different model setups on one grid, so we should not calculate a stable failure probability from them. The concrete lesson is to compare the required case set with the investigated set and independently check the findings. A record of successful tool calls can still hide missing engineering coverage.

Show / explain

Read across the coverage chart by required finding. Point to the missed infeasible outage and ask whether a success-only summary would expose it.

Ask

What evidence would reveal that a critical case was never studied?

Expected answer

The expected-case list, actual returned case IDs and an independent reference or coverage check.

Transition

We also need counts and uncertainty when repeated attempts are available.

Sources: [R01]. Full references follow the glossary.

Open slide 61 online

Slide 62. Show both trial counts and uncertainty

02:11:00-02:13:30 / 2.5 minutes

Say

The capability study reports ten attempts for each shown setup on synthetic 100- and 1,000-bus grids. Six cells have ten successes out of ten and two have five out of ten. Our plotted Wilson intervals illustrate uncertainty under an independent pass/fail model at fixed settings. The paper reports the counts, rather than these calculated intervals. That statistical assumption is not established for every agent workflow. Some higher-complexity tasks and larger grids have no reported result cells in the tested version. Missing evidence should stay missing, rather than become a zero-valued result. Later software fixes also need a new set of measurements before they can support a new numerical performance claim.

Show / explain

Read the counts and whiskers separately. Identify blank or unreported conditions explicitly. Explain that the intervals are tutorial calculations rather than author-provided measurements.

Ask

Is an unreported result equivalent to zero successes?

Expected answer

No. It is missing evidence. Zero successes would require an actual stated set of failed trials.

Transition

Not all benchmarks solve equations. Some test whether agents can assemble the right evidence.

Sources: [R02], [R03]. Full references follow the glossary.

Open slide 62 online

Slide 63. PowerBench finds answers across linked records

02:13:30-02:15:30 / 2 minutes

Say

PowerBench tests retrieval and reasoning over synthetic asset records, documents and time-series data. It has 300 questions in three groups of 60, 90 and 150. The generated collection covers 761 devices and 24,939 documents. Tools search, read and run Python under budgets of 50 tool calls and 300 seconds per attempt. An answer may look well formed yet refer to the wrong asset or time period, making this relevant to evidence work. However, the test does not run power flow or validate operating actions. At the recorded review, the linked repository did not provide a verified runnable release. Explain the benchmark's design without promising an immediately reproducible lab.

Show / explain

Follow the source figure from the linked records into question answering. Point out the identity and time relationships that a correct answer must preserve.

Ask

What kind of operator task is closest to this benchmark?

Expected answer

Finding and combining evidence about assets, events or time periods, rather than calculating physical grid behavior.

Transition

Its different scoring rules answer different questions about those answers.

Sources: [P02]. Full references follow the glossary.

Open slide 63 online

Slide 64. PowerBench results: complete answers and partial credit

02:15:30-02:17:30 / 2 minutes

Say

Joint accuracy requires the complete answer to have every required field and the correct format. Field accuracy gives partial credit when some fields are correct. Macro joint accuracy gives each of the three task groups equal weight. These measures can differ even for the same model responses. The chart's intervals are the authors' bootstrap intervals, obtained by resampling their data. They describe the reported metric and sampling method. Overlap between two separate model intervals does not by itself decide whether a paired model difference is statistically significant. For teaching, focus first on whether the metric matches the operator's need for a complete evidence packet or for partially useful fields.

Show / explain

Identify the joint-accuracy and field-accuracy marks using the legend. Read the axis as a reported score and the whiskers as uncertainty for that score.

Ask

Can high field accuracy coexist with incomplete answers?

Expected answer

Yes. Many individual fields may be correct while too few answers satisfy every required field together.

Transition

The group breakdown makes another weakness visible.

Sources: [P02]. Full references follow the glossary.

Open slide 64 online

Slide 65. A strong average can hide a weak task group

02:17:30-02:19:30 / 2 minutes

Say

The leading row reports joint accuracies of 85.0, 88.9 and 48.7 percent in the three task groups. Their equal-weight average is about 74.2 percent. The largest group, Comparative Retrieval and Reasoning, has fewer than half of its answers completely correct. Equal weighting gives the 60-question group the same influence as the 150-question group. That is a deliberate metric choice, but it answers a different question from pooling all 300 questions. In a pilot, identify which task group matters to the operator and show that group directly. A strong average cannot stand in for acceptable performance on every required task.

Show / explain

Use the source task-family figure to name the three kinds of question. Say each group accuracy before the average, then emphasize the comparative group's lower value.

Ask

Would an average of 74.2 percent establish acceptable performance on the comparative task?

Expected answer

No. That group's reported joint accuracy is 48.7 percent and needs to be judged separately.

Transition

Small trial counts create a further limit even when the observed fraction is perfect.

Sources: [P02]. Full references follow the glossary.

Open slide 65 online

7. Statistics, integration and evaluation

Slide 66. What do ten successful trials tell us?

02:19:30-02:21:30 / 2 minutes

Say

Ten successes out of ten gives an observed success fraction of 100 percent. Under a fixed-setting independent-trial model, the calculated two-sided Wilson 95 percent interval is about 72.2 to 100 percent. Five successes out of ten gives about 23.7 to 76.3 percent. These wide intervals show how little precision a small sample provides. In repeated comparable experiments, the method aims to cover the fixed success probability 95 percent of the time. It does not assign a 95 percent probability to a fixed parameter after this one interval has been observed. Different tasks, shared errors and changed settings create additional uncertainty beyond that model.

Show / explain

Read the two count-and-interval examples slowly. Emphasize the denominator of ten and distinguish a reported fraction from its uncertainty under assumptions.

Ask

Does a 10 out of 10 result show the true success probability is exactly one?

Expected answer

No. It is a small observed sample, and the calculation still depends on the trial model and tested population.

Transition

A multi-step task also depends on how its steps succeed together.

Sources: [R03]. Full references follow the glossary.

Open slide 66 online

Slide 67. More required steps can lower overall reliability

02:21:30-02:23:30 / 2 minutes

Say

Suppose each of 20 required steps succeeds independently with probability 0.99. The chance that all succeed is 0.99 raised to 20, about 81.8 percent. The curve illustrates why strong individual steps can still produce a weaker complete workflow. Real errors may share a cause, such as the wrong input snapshot, cache or adapter. In that case the simple independent-step formula does not describe the system. The exact probability chain uses conditional probabilities. Retrying a step may also repeat the same error rather than provide a fresh independent chance. Measure complete tasks and inspect failure stages instead of predicting a utility's reliability from assumed per-step rates.

Show / explain

Follow the 99-percent curve to 20 steps. Explain that all required steps must succeed for this example. State the independence assumption before using the result.

Ask

Why might a second attempt fail for the same reason as the first?

Expected answer

It may reuse the same wrong context, stale result or defective tool adapter.

Transition

What we count in the final evaluation matters just as much.

Original teaching explanation or worked example. Preserve its stated assumptions.

Open slide 67 online

Slide 68. The answer changes when we change what we count

02:23:30-02:26:00 / 2.5 minutes

Say

This invented example has 80 feasible and 20 infeasible cases. The system accepts 70 feasible cases and rejects ten. It also wrongly accepts two infeasible cases and correctly rejects 18. Among all 100 cases, two false approvals are two percent. Among the 20 infeasible cases, the false-approval rate is ten percent. Among the 72 accepted cases, 70 are feasible, or 97.2 percent. Overall classification accuracy is 88 percent. Each number is arithmetically valid, but each answers a different question. For an operator, false approval of an infeasible case may matter differently from unnecessary rejection of a feasible one. Define those errors before selecting the headline metric.

Show / explain

Walk through all four outcome counts in the graphic. For each percentage, name the denominator aloud. Ask learners which error they would investigate first for their chosen task.

Ask

What is the denominator for the false-approval rate among infeasible cases?

Expected answer

Twenty. Two wrongly accepted infeasible cases out of 20 gives ten percent.

Transition

Even seeing no failures needs a careful interpretation.

Original teaching explanation or worked example. Preserve its stated assumptions.

Open slide 68 online

Slide 69. Zero test failures do not prove perfect performance

02:26:00-02:28:30 / 2.5 minutes

Say

If a fixed system fails independently with probability p on each comparable trial, the chance of seeing zero failures in n trials is one minus p, raised to n. Set that chance to 0.05 and solve for p. The result is the exact one-sided 95 percent upper confidence bound: one minus 0.05 raised to one over n. With 300 zero-failure trials, the bound is about 0.994 percent. The quick rule of three gives a nearby estimate, three divided by n. This is an inference under specific assumptions, rather than certification. It does not cover a different task mix, changed topology, model update or a different tool implementation.

Show / explain

Read the curve as the bound decreasing with more trials. Work through the zero-failure probability in words before naming the formula. Distinguish this one-sided bound from the earlier two-sided Wilson interval.

Ask

Does zero failures in 300 trials prove zero future failures?

Expected answer

No. Under the stated assumptions it yields a nonzero upper bound, and changes outside that population are untested.

Transition

We can design evaluations that make those boundaries visible.

Sources: [R04]. Full references follow the glossary.

Open slide 69 online

Slide 70. Choose tests that can reveal failures

02:28:30-02:30:30 / 2 minutes

Say

A useful test set contains easy and hard tasks, multiple grid sizes, fresh and old inputs, tool errors and both feasible and infeasible cases. Each trial needs a case ID, settings, input snapshot, tool evidence, final status and human corrections. Use an independent reference appropriate to the engineering question. Asking the same model whether its own answer is right does not supply an independent check. Keep final test cases separate from the examples used to tune prompts and adapters. Repeating one prompt many times also leaves task diversity untested. This is a proposed evaluation method. Its value comes from exposing failure types and coverage, rather than producing a single impressive average.

Show / explain

Organize the slide verbally into cases, reference and records. Ask each learner to name one adversarial or difficult condition for their original task.

Ask

Why hold back a final test set during tuning?

Expected answer

To avoid measuring performance on cases that shaped the system and overstating how well it handles unseen work.

Transition

A standard tool interface can help integration, while leaving these tests necessary.

Sources: [P01], [P02], [R02], [E04]. Full references follow the glossary.

Open slide 70 online

Slide 71. MCP connects a workflow to its tools

02:30:30-02:32:30 / 2 minutes

Say

The Model Context Protocol standardizes how an application connects to services exposing tools. The application hosts clients, and each client connects to a server. The server can describe an available operation, accept a request and return a result. In a power workflow, that operation might query a record or call a numerical service. MCP supplies the communication structure. Planning, electrical modeling and validation remain separate responsibilities. The application still checks who is requesting a call, whether the action is allowed, which inputs it uses and what the returned evidence means. An MCP architecture diagram cannot establish numerical accuracy or identify an unreported orchestration library.

Show / explain

Follow host, client and server in the original protocol diagram, then return with the tool result. Give one read-only data call and one simulation call as teaching examples.

Ask

Who decides whether the returned solver evidence answers the user's question?

Expected answer

The application and its engineering review process must apply the task's checks. The protocol alone cannot decide that.

Transition

Documents arriving through those services must also stay within their authority.

Sources: [LG01], [T08], [T09]. Full references follow the glossary.

Open slide 71 online

Slide 72. A retrieved document cannot give itself permission

02:32:30-02:34:30 / 2 minutes

Say

Imagine a retrieved document containing text that tells the assistant to ignore its limits and issue a command. That text is part of the evidence being read. It cannot grant itself operating permission. The application must keep task instructions, user identity and service permissions separate from retrieved content. The trust-boundary diagram shows where the document enters and where an independent action check must apply. Tests should ask whether hostile content changes the case scope, triggers a forbidden call or contaminates the report. Save the blocked request and its reason so the behavior can be inspected. This hostile instruction is an invented classroom example, rather than a reported utility incident.

Show / explain

Trace the retrieved text to the boundary and then to the separate permission check. Ask learners which part of the system can authorize an action.

Ask

Can a manual or webpage legitimately expand the user's operating permissions?

Expected answer

No. Retrieved content is evidence. Authorization must come from the appropriate user and enforced service policy.

Transition

A useful assistant must also improve the complete human task.

Sources: [T09]. Full references follow the glossary.

Open slide 72 online

Slide 73. Measure the whole job, including human review

02:34:30-02:36:30 / 2 minutes

Say

Measure the job from data preparation through agent work, human review and rework. For sequential stages, adding those times gives a useful accounting identity. If work happens in parallel, measure the path that determines completion rather than adding every worker's elapsed time. Include failed attempts, escalations and corrections. A faster draft may still create more reviewer work or miss the operating deadline. Report the spread of times and the difficult cases as well as the average. The conventional monitoring screenshot illustrates the interpretation work an operator does. It is not evidence that the tutorial's timing method was measured in an agent deployment at CAISO.

Show / explain

Point to information the operator must interpret in the real screenshot. Verbally separate machine time, operator attention and total completion time.

Ask

What should we add to a model response-time benchmark before claiming operator time savings?

Expected answer

Preparation, review, correction, failed attempts and an appropriate comparison on completed tasks.

Transition

A pilot can compare that work before authority is expanded.

Sources: [Q01], [U01], [U04], [O05]. Full references follow the glossary.

Open slide 73 online

Slide 74. Test whether the assistant helps on comparable tasks

02:36:30-02:38:30 / 2 minutes

Say

Start with shadow mode: save the assistant's proposals while the existing operating process continues to make the decision. Compare comparable tasks with and without assistance. Task difficulty, practice and learning can affect the result, so balance or randomize order where practical. After tuning, retain a fixed final test set. Define stopping rules, error categories and review responsibilities before expanding the pilot. Record disagreements with independent checks and with reviewers. The photograph shows a real training facility, while the comparison design is our teaching proposal. A pilot supports conclusions about the actual tasks, tools, settings and participants tested. Expansion requires evidence for the new scope.

Show / explain

Use the training-facility photo to explain rehearsing a task under controlled conditions. Connect the proposed comparison to the audience task selected at the start.

Ask

Why can an unbalanced before-and-after comparison exaggerate benefit?

Expected answer

People may have learned the task, conditions may differ or the second set may simply be easier.

Transition

Testbeds provide controlled places for that evaluation.

Sources: [O03]. Full references follow the glossary.

Open slide 74 online

Slide 75. Power-sector testbeds provide places to evaluate agents

02:38:30-02:40:00 / 1.5 minutes

Say

A testbed lets researchers observe a system under selected conditions and deliberately introduce errors or state changes. DOE CESER's July 2026 announcement describes LLNL's Stormbreaker evaluation testbed. EPRI's SAFERai.power develops assessment and risk guidance. These efforts address different roles. The photograph here is ORNL's Grid Operations and Analytics Laboratory, used as power-sector context. It is not a photograph of Stormbreaker. A testbed still needs cases that represent the intended task, and a guidance program does not certify this tutorial's prototype. Use such environments to make failures visible and keep the tested scope explicit.

Show / explain

Identify ORNL as the image source. Name DOE/LLNL and EPRI separately so the photograph does not merge them into a single project.

Ask

Does access to a testbed prove an agent is ready for deployment?

Expected answer

No. It provides an environment in which relevant performance and failure evidence can be collected.

Transition

Our own prototype should face the same standard of evidence.

Sources: [C15], [C11], [C12], [I05]. Full references follow the glossary.

Open slide 75 online

8. Prepare for the three-case workshop

Slide 76. GridGuard: what should this prototype prove?

02:40:00-02:42:00 / 2 minutes

Say

The GridGuard prototype tests proposed actions using OpenDSS on a synthetic feeder. The purpose is to learn whether it improves completed tasks while respecting fixed limits. Save each proposal, its input snapshot, the solver result, rejected alternatives and the reviewer decision. Measure correct feasible proposals, false approvals, modeled limit violations, time and reviewer corrections. The prototype cannot issue live distribution-management commands. Its feeder and results support bounded teaching studies, so we should not present them as a deployed utility control system. A plausible proposal is a starting point. The inspectable evidence and evaluation determine whether it was useful.

Show / explain

Read the feeder as a synthetic model and point to the boundary between proposing and numerical testing. Connect each saved item to one of the three validity checks.

Ask

What would make a proposal useful to an operator reviewing it?

Expected answer

Clear assumptions, current inputs, relevant numerical evidence, visible failures or limits and a decision within the reviewer's authority.

Transition

The next tutorial applies this evidence contract to three cases.

Sources: [L01], [L02]. Full references follow the glossary.

Open slide 76 online

Slide 77. Next tutorial: check the proposed action

02:42:00-02:43:00 / 1 minutes

Say

In voltage support, ask whether the proposed action meets the intended model's limits. In restoration, ask what topology and switching assumptions were checked and which load is restored. In cyber response, ask how uncertain or hostile data affect the evidence and permissions. The hands-on tutorial uses one Classroom Labs notebook to explore those questions. It separates fresh numerical calculations from saved Llama episodes. Carry forward our three checks: valid data, valid engineering and a justified decision. The workshop then lets you run, change and inspect the cases directly.

Show / explain

Name the three cases and the evidence question for each. Show the linked hands-on presentation and Classroom Labs entry point without starting a new demonstration here.

Ask

What should we carry from this session into every lab result?

Expected answer

Its input identity, actual tool evidence, limits, unresolved issues and the human decision boundary.

Transition

Close by asking learners to apply the same questions to their own work.

Sources: [L02]. Full references follow the glossary.

Open slide 77 online

Slide 78. Discussion and full sources

02:43:00-02:45:00 / 2 minutes

Say

Return to the task you chose at the beginning. Which paper or workflow is closest to it? Identify one permitted agent action, one tool that would calculate or retrieve the needed evidence and one result an operator would need to review. Then state a failure that should stop or limit the conclusion. Our central lesson is that a credible claim names the task, the test population, its assumptions, observed failures and remaining human work. The source guide links the original papers, utility reports, documentation and figure credits so you can read beyond the slide. Both exercises have worked answers in this script. The follow-on workshop turns this review method into numerical experiments.

Show / explain

Allow one brief audience example and one question. Return to the opening distinction between reported use and measured benefit. Share the source-guide and hands-on links.

Ask

What evidence would you request before adopting an assistant for your own task?

Expected answer

A bounded assignment, relevant independent tests, inspectable tool records, measured review effort and clear handling of errors and permissions.

Transition

Move to the hands-on tutorial or close the session with the full reading guide.

Original teaching explanation or worked example. Preserve its stated assumptions.

Open slide 78 online

Plain-language glossary

AI agent

A system that observes an environment and chooses actions. A software agent can observe records and act through software interfaces.

Agentic system

Here, a system that uses feedback to pursue a goal. The cited engineering convention includes both programmed workflows and LLM-directed agents.

Generative AI and LLM

Generative AI produces synthetic content. A large language model handles language and can support content creation or decisions within an agent system.

Tool and tool call

A tool performs a defined operation. A tool call is a request with specified inputs. The solver or service returns evidence for the agent to use.

Workflow and orchestration

The process and software rules that coordinate steps, routing, saved state and stopping. Some choices can be fixed in code and some assigned to a model.

Retrieval and RAG

Retrieval finds records. Retrieval-augmented generation supplies selected source text to a language model when it constructs an answer.

DAG

A directed acyclic graph. Arrows show required dependencies and cannot form a directed loop. A task DAG is different from the electrical network.

LangGraph and state

LangGraph runs stateful workflows. State holds information used by later steps. The application defines what the stored fields mean and how updates combine.

Reducer

An application-defined rule for combining state updates. In a parallel study it should preserve result identity and reveal missing or duplicate cases.

Checkpoint, thread and interrupt

A checkpoint saves progress, a thread ID identifies the same run, and an interrupt pauses for outside input. Resumption still needs current-input checks.

Idempotency

Handling a repeated request without unintentionally applying its effect again. This matters when a paused node restarts.

Cache and input fingerprint

A cache stores results for reuse. A key or content hash must identify relevant inputs. Matching an incomplete key can still return a wrong result.

MCP

Model Context Protocol: a standard connection between applications and tool servers. It does not choose studies or prove numerical correctness.

Snapshot, topology and model version

The data time, electrical connection pattern and model revision that define a study. A result should identify all relevant inputs.

Bus and feeder

A bus is a connection point in a power-system model. A feeder is a local distribution network. An electrical edge has a different meaning from a workflow edge.

Power flow

A numerical calculation of a steady operating condition for specified inputs. Its result does not establish dynamic stability.

OPF and ACOPF

Optimal power flow chooses allowed variables while meeting modeled constraints. ACOPF uses alternating-current network equations.

Dynamic study

A simulation of system behavior over time, including equipment and controller response. Conclusions depend on the model and events tested.

N-1 contingency

A study of a selected single-component outage. A completed selected list is not automatically every relevant N-1 case.

Dispatch and unit commitment

Dispatch sets output of selected units. Unit commitment changes which units are on. Changing both enlarges a dispatch-only decision space.

Reactive power and per unit

Reactive power is part of an AC study, commonly expressed in var or Mvar. Per unit is a value divided by a compatible base quantity.

Solver residual and tolerance

A residual measures remaining equation error. A tolerance specifies accepted numerical error. These checks need the correct problem and inputs.

Nonconvergence and infeasibility

Nonconvergence means the solve did not obtain an acceptable result. It alone does not prove no feasible solution exists.

Surrogate and reduced-order model

An approximation that can reduce computational work. Its validity must be checked for the intended task and conditions.

Pass@1

Success on the first evaluated attempt under a specified scoring rule. Its meaning depends on the benchmark and is not a generic safety measure.

Joint, field and macro accuracy

Joint accuracy requires the complete answer. Field accuracy gives partial credit. A macro average gives groups equal weight rather than pooling all cases.

Confidence interval

An interval procedure with stated repeated-sample coverage under assumptions. It is not proof of performance on changed tasks or settings.

False approval

Accepting a case that should fail the stated criteria. Always name the denominator, such as all infeasible cases or all accepted cases.

Shadow mode

Saving assistant proposals while the existing decision process continues. It allows comparison before the assistant receives greater authority.

Data, engineering and decision validity

Respectively: correct evidence identity and consistency; the required numerical checks; and sufficient evidence and authority for the next decision.

Full references and source credits

These 76 entries follow the accepted deck. The source figures retain their original attribution. The full reading guide includes detailed visual provenance and the original evidence limits.

Full reading guide

[B01] IEEE Communications Society. IEEE Communications Society: official resources and network presentation template. Reviewed 17 October 2026.

https://www.comsoc.org/about/brand-experience/resources

[DEF03] Erik S. and Barry Zhang. Building effective agents. Anthropic Engineering, December 19, 2024. Live article reviewed October 17, 2026.

https://www.anthropic.com/engineering/building-effective-agents

[DEF02] Wooldridge, Michael, and Nicholas R. Jennings. Intelligent agents: theory and practice. The Knowledge Engineering Review 10, no. 2 (June 1995): 115-152. https://doi.org/10.1017/S0269888900008122.

https://doi.org/10.1017/S0269888900008122

Full paper: https://www.cs.ox.ac.uk/people/michael.wooldridge/pubs/ker95.pdf

[NV01] Eduardo Alvarez, Benjamin Klieger, Graham Steele. Building for the Rising Complexity of Agentic Systems with Extreme Co-Design. NVIDIA Technical Blog. May 5, 2026.

https://developer.nvidia.com/blog/building-for-the-rising-complexity-of-agentic-systems-with-extreme-co-design/

[GEN01] National Institute of Standards and Technology. Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile. NIST AI 600-1. July 2024.

https://doi.org/10.6028/NIST.AI.600-1

Full paper: https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf

[DEF01] Russell, Stuart J., and Peter Norvig. Artificial Intelligence: A Modern Approach. 2nd ed. Upper Saddle River, NJ: Prentice Hall, 2003. Chapter 2, Intelligent Agents, pp. 32-58. Author-hosted chapter draft dated 2002.

https://aima.cs.berkeley.edu/2nd-ed/

Full paper: https://aima.cs.berkeley.edu/4th-ed/pdfs/newchap02.pdf

[DEF06] LangChain maintainers. Tools. Docs by LangChain, Python open-source documentation. Accessed 17 October 2026.

https://docs.langchain.com/oss/python/langchain/tools

[NV02] Tanay Varshney. Introduction to LLM Agents. NVIDIA Technical Blog. November 30, 2023.

https://developer.nvidia.com/blog/introduction-to-llm-agents/

[DEF04] LangChain maintainers. LangGraph overview and Tools. Docs by LangChain, Python open-source documentation. Accessed October 17, 2026.

https://docs.langchain.com/oss/python/langgraph/overview

[LG04] LangChain documentation maintainers. Workflows and agents. Docs by LangChain, Python LangGraph documentation. Undated live documentation; accessed October 18, 2026.

https://docs.langchain.com/oss/python/langgraph/workflows-agents

[LG03] LangChain documentation maintainers. LangGraph Interrupts. Official LangChain documentation. Accessed 2026-10-17; live documentation.

https://docs.langchain.com/oss/python/langgraph/interrupts

[PA02] Power and AI Initiative / PowerAgent community, Harvard SEAS. PowerAgent - Open-Source Agentic AI for Power Grids. Official Harvard project website. Live project page reviewed 17 October 2026.

https://poweragent.seas.harvard.edu/

[PD01] Emmanuel O. Badmus, Amritanshu Pandey. PowerDAG: Supervisory Agentic AI System for Automating Distribution Grid Analysis. arXiv preprint, version 4. 5 July 2026 (v4); first submitted 18 March 2026.

https://arxiv.org/pdf/2603.17418v4

[N01] Tapestry. Field Notes HyperQ deployment in PJM's Cycle 1. 2026-06-11; deployment entry 2026-04-29.

https://www.tapestryenergy.com/en/projects/field-notes-hyperq-deployment-in-pjm-s-cycle-1

[N02] Tapestry. How agentic AI can help grid operators speed up interconnection. 2025-12-08 (linked chronological entry in N01); live article reviewed 2026-10-17.

https://www.tapestryenergy.com/en/projects/how-agentic-ai-can-help-grid-operators-speed-up-interconnection

[N06] PJM Interconnection. Successful Interconnection Reforms, Other PJM Initiatives Seek To Maximize Electricity Supplies. 2026-03-16.

https://insidelines.pjm.com/connected/

[C01] California ISO. Exploring AI to support smarter grid management. 2025-07-28.

https://www.caiso.com/about/news/energy-matters-blog/exploring-ai-to-support-smarter-grid-management

[C02] OATI. Generative and Agentic AI in Grid Operations: Field Lessons from OATI, CAISO, PNNL, and EPRI. 2026 (exact publication day not displayed).

https://www.oati.com/ai-for-energy/

[I06] OATI (California ISO control-room photograph). California ISO control room, image published by OATI: California ISO control room with operators seated at curved desks facing a long wall of grid displays. Image accessed 17 October 2026.

https://www.oati.com/ai-genie/

[U02] Amazon Web Services. AWS Launches Agentic Grid Planning Program to Accelerate Interconnection Studies. 2026-09-17.

https://press.aboutamazon.com/aws/2026/9/aws-launches-agentic-grid-planning-program-to-accelerate-interconnection-studies

[U03] Amazon Web Services. Agentic Grid Planning on AWS. Undated; reviewed 17 October 2026.

https://aws.amazon.com/energy-utilities/agentic-grid-planning/

[I03] Duke Energy / illumination. Duke Energy, illumination, August 2, 2022: Duke Energy grid operator Howard Greene seated before many control-room displays. Image accessed 17 October 2026.

https://illumination.duke-energy.com/articles/meet-the-team-that-monitors-the-grid-and-keeps-the-lights-on

[N03] Siemens. Siemens' Gridscale X redefines system operations and agentic transmission planning. 2026-05-05.

https://press.siemens.com/global/en/pressrelease/siemens-gridscale-x-redefines-system-operations-and-agentic-transmission-planning

[N04] National Grid. National Grid Full Year Results 2025/26: Video Webcast Transcript. 2026-05-14.

https://www.nationalgrid.com/document/577036/download

[N09] Session speakers: Saptaparna Gupta (NYPA), Oleksandra Bovkun (Databricks), Eliza Pochinova (Ataccama). On demand: Trust-driven medallion architecture: How to make your CDEs ready for AI. Ataccama. 2026-10-14 (on-demand page); event page currently lists 2026-10-13.

https://www.ataccama.com/webinar/on-demand-trust-driven-medallion-architecture

[N10] Saptaparna Gupta (NYPA), Oleksandra Bovkun (Databricks), Eliza Pochinova (Ataccama). Trust-driven medallion architecture: how to make your CDEs ready for AI. Ataccama. 2026-10-13, 11:00 AM EDT (date displayed on fetched event page).

https://www.ataccama.com/events/trust-driven-medallion-architecture

[N07] Eversource Energy (employer). Distribution System Modeling and Automation Engineer (Hybrid Schedule). Eversource Energy. 2026-10-05.

https://jobs.eversource.com/job/east-berlin/distribution-system-modeling-and-automation-engineer-hybrid-schedule/7774/10157 6695504

[N11] MathWorks; named Eversource engineer John Kreso quoted. Eversource Energy Collaborates with MathWorks to Incorporate Renewable Energy Sources into System Planning Process Using Probabilistic Load Flow Automation. MathWorks. 2025-04-29.

https://www.mathworks.com/company/newsroom/eversource-energy-collaborates-with-mathworks-to-incorporate-renewable-energy-sources-into-system-planning-process-using-probabilistic-load-flow-automation.html

[C07] GE Vernova. GE Vernova Launches GridOS for Distribution, the Industry's First Unified Solution for Grid Orchestration. 2026-02-03.

https://www.gevernova.com/news/press-releases/ge-vernova-launches-gridos-distribution-industrys

[I01] GE Vernova. GE Vernova: GE Vernova GridOS promotional control-room composite with people and grid software displays. Image accessed 17 October 2026.

https://www.gevernova.com/software/products/gridos

[Q01] Corey Fallon, Brett Jefferson, Tyler Ryan. An Analysis of Grid Operators. Pacific Northwest National Laboratory. 2023-04-19.

https://www.pnnl.gov/sites/default/files/media/file/PNNL_workshop_Presentation_Human_Factors_Virtual_Workshop_March_2023.pdf

[O01] PJM Interconnection. What Happens in the Control Room. 2026-04-16.

https://www.pjm.com/-/media/DotCom/about-pjm/newsroom/fact-sheets/what-happens-in-the-control-room-fact-sheet.pdf

[I02] Duke Energy / illumination. Duke Energy, illumination, August 2, 2022: Duke Energy employees in an Energy Control Center, with one pointing toward a display. Image accessed 17 October 2026.

https://illumination.duke-energy.com/articles/meet-the-team-that-monitors-the-grid-and-keeps-the-lights-on

[O05] John Phipps. Control center modernization is enhancing reliability. California ISO. 2025-09-10.

https://www.caiso.com/about/news/energy-matters-blog/control-center-modernization-is-enhancing-reliability

[I09] GE Vernova. GridOS Visual Intelligence product screens. Accessed 17 October 2026.

https://www.gevernova.com/software/products/gridos/visual-intelligence

[LG01] LangChain documentation maintainers. LangGraph Graph API overview. Official LangChain documentation. Accessed 2026-10-17; live documentation.

https://docs.langchain.com/oss/python/langgraph/graph-api

[E01] Yihan (Logon) Wen, Xin Chen. X-GridAgent: An LLM-Powered Agentic AI System for Assisting Power Grid Analysis. arXiv preprint. 2025-12-23 (v1).

https://arxiv.org/pdf/2512.20789v1

[R01] Samim Konjicija, Slaven Peles. Integrating Agentic Artificial Intelligence with High-Performance Computing for Grid Planning. arXiv; EnergyCon 2026 proceedings forthcoming as described by authors. 3 September 2026; v1.

https://arxiv.org/abs/2609.04544v1

Full paper: https://arxiv.org/pdf/2609.04544v1

[L01] GridGuard project / local source snapshot. GridGuard current project: README and graph source snapshot. Snapshot 17 October 2026; see research/GridGuard_source_snapshot.txt.

https://alaaselim-sudo.github.io/GridGuard-Agentic-AI-Tutorials/foundations/research/GridGuard_source_snapshot.txt

[D01] Barun Kumar Saha, Aarthi V, O. D. Naidu. DrAgent: An Agentic Approach to Fault Analysis in Power Grids Using Large Language Models. 2025 International Conference on Artificial Intelligence in Information and Communication (ICAIIC), pp. 938-945. 2025-02-18-21.

https://www.manuscriptlink.com/society/kics/media?key=kics%2Fconference%2Ficaiic2025%2Fpresentation%2F1571098105.pdf

[E02] Mengshuo Jia, Zeyu Cui, Gabriela Hug. Enhancing LLMs for Power System Simulations: A Feedback-Driven Multi-Agent Framework. IEEE Transactions on Smart Grid, volume 16, issue 6, pages 5556-5572. Published online 2025-07-18; issue November 2025; inspected full arXiv v3 dated 2025-05-19.

https://doi.org/10.1109/TSG.2025.3589114

Full paper: https://arxiv.org/pdf/2411.16707v3

[E03] Yan Zhang, Ahmad Mohammad Saber, Amr Youssef, Deepa Kundur. Grid-Agent: An LLM-Powered Multi-Agent System for Power Grid Control. arXiv preprint. 2025-09-08 (v3); first submitted 2025-08-07.

https://arxiv.org/pdf/2508.05702v3

[P02] Xijing Wang, Yinsheng Yao, Jinru Ding, Yidong Jiang, Ziwen Xu, Yiwen Jiang, Jie Xu, Dawei Cheng. PowerBench: A Benchmark for Agentic Retrieval and Reasoning in Power Systems. arXiv. 2026-09-28 (v1).

https://arxiv.org/abs/2609.34492v1

Full paper: https://arxiv.org/pdf/2609.34492v1

[I04] NVIDIA. Image published by NVIDIA with its EPRI Open Power AI Consortium announcement, March 20, 2025: Transmission towers and power lines with city lights in the background at dusk. Image accessed 17 October 2026.

https://blogs.nvidia.com/blog/open-power-ai-consortium/

[PA01] Qian Zhang, Le Xie. PowerAgent: A Road Map Toward Agentic Intelligence in Power Systems: Foundation Model, Model Context Protocol, and Workflow. IEEE Power and Energy Magazine, volume 23, issue 5, pages 93-101. September-October 2025 issue; TechRxiv preprint DOI version 1, June 2025.

https://doi.org/10.1109/MPE.2025.3579718

[PA03] Le Xie; slides credit Le Xie and Qian Zhang. A Roadmap Towards Agentic Intelligence in Power Systems. ERCOT Technology and Security Committee guest presentation. 8 December 2025.

https://www.ercot.com/files/docs/2025/12/01/4-Guest-Presentation-A-Roadmap-toward-Agentic-Intelligence-in-Power-Systems.pdf

[P01] Qian Zhang, Andrea Pomarico, Costas Mylonas, Magda Foti, Alberto Berizzi, Le Xie. PowerAgentBench-Dyn: A Benchmark for Agentic AI in Power System Dynamic Studies. arXiv. 2026-06-18 (v1).

https://arxiv.org/abs/2606.20401v1

Full paper: https://arxiv.org/pdf/2606.20401v1

[PA06] PowerAgent / PowerWF community contributors. PowerWF Grid Impact Evaluation - LangGraph workflow source. Author-maintained source code, Grid Impact Evaluation/src/agent/graph.py. Live source inspected 17 October 2026.

https://github.com/Power-Agent/PowerWF/blob/main/Grid%20Impact%20Evaluation/src/agent/graph.py

[PA05] PowerAgent community contributors. Power-Agent/PowerWF. Author-maintained GitHub repository. Live source repository reviewed 17 October 2026.

https://github.com/Power-Agent/PowerWF

[P03] D. Adrian Maldonado, Jonghwan Kwon, Sunash B. Sharma, Hongwei Jin, Stefano Fenu, Emil Constantinescu, David Sehloff, Parfait Gasana, Duane Rudolph Verner, Kibaek Kim. GridMind: An Agentic Workflow for Weather-Driven Power Grid Risk Assessment. KDD 2026 Workshop on Scientific and Societal Agents / OpenReview. August 2026 workshop; reviewed 17 October 2026.

https://openreview.net/forum?id=Cs5TBzzNym

Full paper: https://openreview.net/pdf/5ad1c2afd7298f9e6ce3bc25f18e7165555e1ffd.pdf

[LG02] LangChain documentation maintainers. LangGraph Persistence. Official LangChain documentation. Accessed 2026-10-17; live documentation.

https://docs.langchain.com/oss/python/langgraph/persistence

[LG05] LangChain documentation maintainers. Checkpointers. Docs by LangChain, Python LangGraph documentation. Undated live documentation; accessed October 18, 2026.

https://docs.langchain.com/oss/python/langgraph/checkpointers

[PD02] Emmanuel O. Badmus and contributors. DistGrid-AgentBench: A Distribution Grid Benchmark for Agentic AI. Author-maintained GitHub repository. Live repository reviewed 17 October 2026.

https://github.com/emmanuelbadmus/DistGrid-AgentBench

[R02] Eve Tsybina, Samim Konjicija, Slaven Peles. Agentic Artificial Intelligence for Power Systems: Strategies to Identify and Close Capability Gaps. arXiv. 30 July 2026; v1.

https://arxiv.org/abs/2607.28822v1

Full paper: https://arxiv.org/pdf/2607.28822v1

[VID01] LangChain. LangSmith Studio: The first agent IDE. Launch walkthrough linked 1 August 2024.

https://www.youtube.com/watch?v=pLPJoFvq4_M

[VID02] Siemens. Gridscale X PSS E: agentic transmission planning - product video. Undated video; accessed 26 October 2026.

https://video.sw.siemens.com/embed/682ca322-5d0b-47a0-8d18-77bf9d0fb016

[P04] Hongwei Jin, Kibaek Kim, Jonghwan Kwon. GridMind: LLMs-Powered Agents for Power System Analysis and Operations. arXiv; published as SC Workshops 2025, DOI 10.1145/3731599.3767409. 2025-09-02 (arXiv v1); SC Workshops 2025.

https://arxiv.org/abs/2509.02494v1

Full paper: https://arxiv.org/pdf/2509.02494v1

[P03A] Argonne GridAI project team. GridAI - AI Foundation Models & Agents for the Power Grid. Live project page reviewed 17 October 2026.

https://argonne-gridfm.github.io/

[P01C] PowerAgent Community. Power-Agent/PowerAgentBench. Live author repository reviewed 17 October 2026.

https://github.com/Power-Agent/PowerAgentBench

[E04] Shivanshu Tripathi, Hamed Mohsenian-Rad, Maziar Raissi. VeraGrid-Agent: Tool-Augmented LLMs for Distribution Optimal Power Flow at the Grid Edge. arXiv preprint. 2026-07-28 (v1).

https://arxiv.org/pdf/2607.25155v1

[E04C] Shivanshu Tripathi / VeraGrid-Agent authors. Shivanshu-11/veragrid-llm-benchmark. Author GitHub repository. Reviewed 2026-10-17.

https://github.com/Shivanshu-11/veragrid-llm-benchmark

[R03] National Institute of Standards and Technology, SEMATECH. 7.2.4.1. Confidence intervals. NIST/SEMATECH e-Handbook of Statistical Methods. Undated live handbook section; reviewed 17 October 2026.

https://www.itl.nist.gov/div898/handbook/prc/section2/prc241.htm

[R04] National Institute of Standards and Technology. EXACT BINOMIAL. NIST Dataplot Reference Manual. Last updated 16 November 2015; reviewed 17 October 2026.

https://www.itl.nist.gov/div898/software/dataplot/refman2/auxillar/exacbino.htm

[T08] Model Context Protocol. MCP Architecture overview. 2026-07-28 documentation revision.

https://modelcontextprotocol.io/docs/2026-07-28/learn/architecture

[T09] Model Context Protocol. MCP Security Best Practices. 2026-07-28 documentation revision.

https://modelcontextprotocol.io/docs/2026-07-28/tutorials/security/security_best_practices

[U01] Google Cloud. AES: Scaling energy safety audits with Claude on Google Cloud. Undated; reviewed 17 October 2026.

https://cloud.google.com/customers/aes

[U04] Amazon Web Services. TC Energy increases office productivity with automation on Amazon Bedrock. Undated; reviewed 17 October 2026.

https://aws.amazon.com/solutions/case-studies/tcenergy-bedrock/

[O03] Tricia Johnstone, Director, Operational Readiness. ISO now has expanded training and emergency operations facilities. California ISO. 2025-02-20.

https://www.caiso.com/about/news/energy-matters-blog/iso-now-has-expanded-training-and-emergency-operations-facilities

[C15] U.S. Department of Energy. CESER Releases New Testbed to Advance LLM and Agentic AI Evaluation for Critical Infrastructure. 2026-07-16.

https://www.energy.gov/ceser/articles/ceser-releases-new-testbed-advance-llm-and-agentic-ai-evaluation-critical

[C11] EPRI via GlobeNewswire. EPRI Launches Effort to Reduce Risk in AI Power Sector Deployment. 2026-08-06.

https://www.globenewswire.com/news-release/2026/08/06/3340219/0/en/epri-launches-effort-to-reduce-risk-in-ai-power-sector-deployment.html

[C12] EPRI. SAFERai.power. Undated; reviewed 17 October 2026.

https://msites.epri.com/saferai

[I05] Oak Ridge National Laboratory, U.S. Department of Energy. Oak Ridge National Laboratory, U.S. Department of Energy: Researchers at ORNL GOAL facing a large display of U.S. grid and geographic data. Image accessed 17 October 2026.

https://www.ornl.gov/grid-operations-and-analytics-laboratory-goal/

[L02] Local workshop source snapshot. Existing three-use-case GridGuard workshop: instructor route and synthetic-feeder context. Snapshot 17 October 2026.

https://alaaselim-sudo.github.io/GridGuard-Agentic-AI-Tutorials/foundations/research/Follow_on_workshop_source.txt

[P02C] OpenCompass / PowerBench authors. open-compass/PowerBench. Live author-linked repository reviewed 17 October 2026.

https://github.com/open-compass/PowerBench

[P03M] KDD26SciSocLLM workshop organizers. KDD 2026 SciSoc LLM Workshop - Accepted Papers. 2026 workshop listing.

https://github.com/KDD26SciSocLLM/KDD26SciSocLLM.github.io/blob/master/accepted-papers.md

[PA04] PowerAgent community contributors. Power-Agent/PowerMCP. Author-maintained GitHub repository. Live source repository reviewed 17 October 2026.

https://github.com/Power-Agent/PowerMCP